



Agglomeration economies in developing countries: A meta-analysis

Arti Grover, Somik Lall, Jonathan Timmis^{*}

The World Bank Group, 1818H. Street, N.W, Washington DC, 20433, USA

ARTICLE INFO

JEL classification:

R11
R12
R30
O18

Keywords:

Meta-analysis
Agglomeration
Developing countries
Productivity
Elasticity

ABSTRACT

Recent empirical work suggests that agglomeration forces are multiple times higher in developing countries than in advanced economies, but also that these cities are crowded and dysfunctional. To understand the true productivity advantages of developing country cities, we systematically evaluate nearly 1300 agglomeration elasticity estimates from 76 studies in 34 countries. Using frontier methodology for conducting meta-analyses, we find that the elasticity in developing countries are less than 1 percentage point higher than in advanced economies, with the difference not being statistically significant. Further, we present novel estimates of urban costs in developing and developed country cities – pollution, carbon emissions, wellbeing, homicides and congestion. While the levels of urban costs and the elasticity of crime are higher in developing country cities, other measures of urban cost elasticity are not different across income groups.

1. Introduction

There is a growing body of academic literature highlighting the productivity enhancing agglomeration economies from living and working in dense cities (Ahlfeldt and Pietrostefani, 2019). Duranton and Puga (2004) outline the micro-foundations of agglomeration economies based on sharing, matching, and learning mechanisms. Dense cities encourage *sharing* of indivisible public goods, production facilities, and marketplaces, a greater variety of inputs and individual specialization, and the pooling of risk; improve the quality and possibility of *matching* between firms and workers; and increase opportunities for *learning* from the generation, diffusion, and accumulation of knowledge.

Economists have used the elasticity of wages with respect to urban density as a canonical measure of agglomeration economies. A recent meta-analysis of empirical work by Ahlfeldt and Pietrostefani (2019) covering 347 estimates shows a doubling urban density is associated with a wage premium of 4 percent. In the United States, the elasticity of wages with respect to city size is 0.043; in France it is 0.03 - implying that doubling density could increase productivity by 3–4 percent. (Combes and Gobillon 2015; Melo et al., 2009; Rosenthal and Strange 2004).

While these estimates suggest strong productivity enhancing agglomeration economies in dense urban environments, a major knowledge gap arises from the limited knowledge of agglomeration economies in developing country settings. The empirical reviews (Melo

et al., 2009; Ahlfeldt and Pietrostefani, 2019) and a recent literature survey (Duranton and Puga, 2020) primarily draw on developed country estimates. The meta-analysis carried out by Ahlfeldt and Pietrostefani (2019) suggests an elasticity of productivity with respect to density of 0.04, based on a citation-weighted average of 347 estimates. However, there has been no attempt to systematically quantify the differences in estimates between advanced and developing countries. Filling this knowledge gap is important as most of future urban growth is expected in developing country cities (United Nations 2018) and these cities are growing in people but without the commensurate investments in human and physical capital that enhance the returns from density (Lall et al., 2017; Ellis and Roberts 2016).

Avner and Lall (2016) find jobs are often located far from where people live; heavy congestion and high rates of walking and informal transportation fragments the labor market and leads to low employment rates and the misallocation of labor in Nairobi. For instance, *Matatu* (privately owned minibuses) users on average can access only 4 percent of jobs within 30 min, 10 percent within 45 min, and 20 percent within 60 min. In metropolitan Buenos Aires, equivalent accessibility figures using public transportation are only 7 percent, 18 percent, and 34 percent for the same time thresholds (Peralta-Quirós, 2015). In Ugandan cities only 19 percent of the city's jobs are accessible within a 1 h commute (Bernard, 2016). Further, many cities particularly in Sub Saharan Africa and South Asia produce non-tradable goods and services (Lall et al., 2017; Venables, 2017). This is consistent with what has been

^{*} Corresponding author.

E-mail addresses: agrover1@worldbank.org (A. Grover), slall1@worldbank.org (S. Lall), jtimmis@worldbank.org (J. Timmis).

<https://doi.org/10.1016/j.regsciurbeco.2023.103901>

Received 17 March 2022; Received in revised form 28 February 2023; Accepted 6 May 2023

Available online 9 June 2023

0166-0462/© 2023 Published by Elsevier B.V.

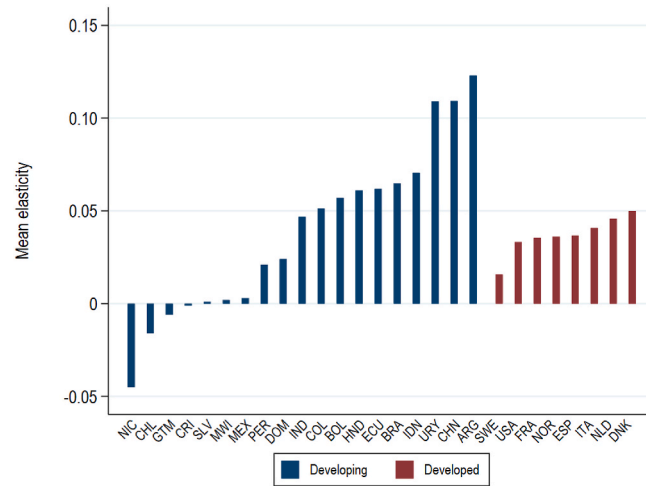


Fig. 1. Nominal wage data show significant agglomeration benefits in developing countries

Notes: The chart reflects the unweighted average wage elasticity for each country using estimates that focus on manufacturing or the whole economy. Developed reflects studies of high-income countries, developing reflects non-high income, with income level determined using the World Bank country-income classification at the mid-year of each study's sample period. The chart reflects 457 raw elasticity estimates, 218 from developed countries and 239 from developing countries. The chart reflects two-thirds of all our developing country estimates that we use in the meta-analysis.

called “pre-mature urbanization” (see for e.g. [Gollin et al., 2016](#)), that is, people are concentrating in developing country cities but not because industrial dynamism is attracting them.

To shed light on the productivity advantages of agglomeration in developing country cities, we make the following contributions. *First*, we expand the sample size of existing meta-analyses that focused on advanced economies to systematically examine 1281 elasticity estimates, originating from 76 studies covering 22 developing and 12 advanced countries. *Second*, compared to studies measuring the benefits of agglomeration, the evidence base for urban costs is generally under-developed. We construct novel estimates of density elasticity on urban costs, with respect to crime, congestion, wellbeing and pollution by collecting data from hundreds of cities around the world, including several in developing countries. *Third*, using frontier methodology for conducting meta-analysis, we control for a variety of differences across studies to provide a robust assessment. This allows us to contextualize country specific estimates showing large agglomeration economies: 0.19 in China, 0.12 in India, 0.17 in Africa, and between 0.06 and 0.16 in Latin America ([Chauvin et al., 2017](#) (India, China), [Combes et al., 2019](#) (China), [Henderson et al., 2019](#) (six African countries: Ethiopia, Nigeria, Ghana, Malawi, Tanzania and Uganda) and [Quintero and Roberts 2018](#) (Latin America)). *Fourth*, we present a back-of-the-envelope calculation of the monetary value of a 1% increase in density in developed and developing country cities using our meta-agglomeration estimates and new urban cost measures.

Our meta-analysis finds that elasticity estimates for developing countries is less than 1 percentage point higher than that for developed countries and the estimates are not statistically different. Nonetheless, estimates using nominal wages, the canonical measure of agglomeration economies, are higher than those using TFP. This suggests that part of the wage premium may be driven by higher capital intensity, perhaps a result of thicker capital markets in urban areas, or higher urban costs, rather than efficiency or spillovers per se.

Further, studies that report real agglomeration premia (accounting for urban costs) find elasticities to be 4.2 percentage points lower than studies that do not, implying a net agglomeration elasticity of essentially zero (−0.6 percent) when using labor productivity as an outcome measure. Our novel estimates of urban dis-amenities suggest that only the elasticity of crime with respect to density is higher in developing country cities. Although the elasticity of pollution, CO2 emissions,

wellbeing and congestion in developing countries is comparable with developed countries, the levels of urban costs are much higher (with the exception of carbon dioxide emissions). For the average city density in our data, in developed countries 28 percent fewer hours are spent in traffic congestion, pollution is 23 percent lower, wellbeing is 15 percent lower, and the homicide rate is around 3 times lower. In contrast, carbon dioxide emissions are 39 percent higher in developed country cities. A back-of-the-envelope calculation suggests that the benefits in terms of higher gross wages in both developed and developing country cities are offset by the higher costs of pollution, congestion and lower subjective wellbeing.

Our analysis confirms the high wage elasticities with respect to density; however, we find no evidence for efficiency gains in light of high and increasing costs of working in developing country cities. This may be driven by bad design and lack of capital investment in cities, but also by the fact that their growth is not driven by the process of structural transformation, which would create a mass of industrial or service firms that benefit from sharing, matching, and learning ([Gollin et al., 2016](#)). Many developing country cities are not dense and productive—they are just crowded.

The rest of the paper is organized as follows. Section 2 reviews the relevant literature for both agglomeration economies and urban costs. Section 3 describes data collected for the meta-analysis and urban cost estimation. Section 4 discusses the estimation strategy, Section 5 presents meta-analysis results on agglomeration elasticities and Section 6 presents the results on urban costs. Section 7 presents monetary equivalents of our estimates and Section 8 concludes.

2. Density, productivity and urban cost elasticities

To set the stage for our subsequent meta-analysis and inform our choice of study characteristics included as meta-controls, following [Gaubert \(2018\)](#) and [Combes and Gobillon \(2015\)](#), we postulate that a firm of efficiency z in sector j and city of size L produces output Y_j according to the following Cobb-Douglas production function:

$$Y_j(z, L) = \varphi(z, L, I_j) k^{\alpha_j} (s \bullet l)^{1-\alpha_j} \quad (1)$$

where l and k denote labor and capital inputs, s is a measure of skills, α_j is the capital intensity of all firms in sector j and $\varphi(z, L, I_j)$ is a firm-specific

Hicks-neutral productivity shifter. The productivity of a firm $\varphi(z, L, I_j)$ increases with its own ‘raw’ efficiency z , a sector-specific parameter I_j and with local agglomeration externalities that depend on city size L .

2.1. Agglomeration measures

City size (L) can be measured using population, density and measures of market potential or access. The use of population or economic density is preferred relative to population mass because spatial units are often based on administrative boundaries and there is large heterogeneity in the size of these units (Ciccone and Hall, 1996). Measures of agglomeration such as market potential account for (economic or physical) distance of the region from other spatial units (Harris, 1954). Market access measures adjust market potential with local price effects to account for imperfect competition across locations (Fujita et al., 1999).

Market access is typically computed by aggregating the income of other municipalities discounted by the distance (that is, some measure of travel cost) to the municipality under consideration. Elasticity estimates with market access measure paint a rather dismal picture on the returns to agglomeration. In Colombia, for example, the elasticity of wages with respect to external market access is significantly negative, in comparison to the estimates with respect to city population, which is about 5 percent, (Duranton, 2016). We do not use estimates using market access in our meta-analysis due to their limited number relative to other agglomeration measures as well as the heterogeneity in their use of travel costs in their construction.

2.2. Productivity measures

Most studies use *nominal* wages as a measure of city productivity. Wages can be recovered using the first order conditions for maximizing (1), where the relative wage (relative to capital rent) is equal to the share relative marginal product of labor and capital:

$$w_j = s \cdot \left(p_j \frac{\varphi(z, L, I_j)}{r_j^{\alpha_j}} \right)^{1/(1-\alpha_j)} \quad (2)$$

Fig. 1 suggests that developing country cities generate huge benefits from agglomeration, measured by nominal wages (see Fig. 1). These are consistent with agglomeration elasticities of 0.19 in China, 0.12 in India, 0.17 in Africa, and between 0.06 and 0.16 in Latin America (Chauvin et al., 2017 (India, China), Combes et al., 2019 (China), Henderson et al., 2019 (six African countries: Ethiopia, Nigeria, Ghana, Malawi, Tanzania and Uganda) and Quintero and Roberts 2018 (Latin America)).

Nominal wages increase not only because of the Hicks-neutral productivity shifter, but also because of worker skill s (which we discuss later) and the as well as the price of goods p_j and the cost of capital inputs r_j (see equation (2)). As emphasized by Combes and Gobillon (2015), better access to markets in cities can increase the price of goods and reduce the cost of inputs, both of which increase nominal wages (we discuss urban costs below). Therefore, *real* wages may be a natural indicator of city wage premia, however the main challenge is the availability of price data at a local level. Even if prices are available, it is not clear whether the estimated agglomeration economies with respect to real wages represent sorting on human capital or compensation for adverse urban amenities rather than true productivity premia of cities (Chauvin et al., 2017). Wages are usually only proportional to and not equal to labor productivity by a factor that depends on the local monopsony power of the firm.

Instead, with firm-level data on capital, one can use measures of total factor productivity (TFP). TFP can be recovered from (1), where we assume a value-added production function¹:

$$\ln TFP_j = \ln p_j Y_j - \alpha_j \ln k - (1 - \alpha_j) \ln l = \ln p_j \varphi(z, L, I_j) s^{1-\alpha_j} \quad (3)$$

TFP depends on the Hicks-neutral productivity shifter, local output prices and skilled labor. Unlike wage estimation, TFP does not depend on the cost of inputs r_j . Thus, the use of total factor productivity (TFP) is preferred since urban costs do not play a role and it avoids making any assumption about the relationship between the local monopsony power and agglomeration economies (Combes and Gobillon, 2015).

Melo et al. (2009) show that elasticities of TFP with respect to density are on average estimated to be larger than those obtained for wages, typically around 50 percent larger. In France, for example, the elasticity of TFP with respect to density of 0.035–0.040 whereas with the same data, the elasticity is 0.027 for wages (Combes et al., 2012). It is difficult to interpret the difference between the two types of estimates. In wage equations, all the effects are re-scaled by the share of labor in the production function. Moreover, agglomeration economies percolating through the cost of inputs other than labor, such as land and intermediate inputs affect wages, but not TFP. A further possible reason for the difference in estimates obtained from wage and TFP regressions is that most researchers have not managed to successfully control for worker skills in wage regressions.

2.3. Endogeneity – omitted local factors

A regression of productivity on a measure of agglomeration size is likely to be biased because of omitted variables – so-called endogeneity issues. Unobservables that affect both density and productivity could include local infrastructure (e.g. roads, railways, airports) or provision of local amenities (e.g. schools or universities).

The literature suggests two approaches to address this endogeneity problem: (i) instrumental variable estimations using historical measures of density (Ciccone and Hall 1996),² geological variables such as land fertility (Combes et al., 2010), land suitability for the construction of tall buildings (Rosenthal et al., 2008; Combes et al., 2010). (ii) quasi-experimental designs such as the location of the Berlin Wall (Ahlfeldt et al., 2015) (iii) including fixed effects to capture any unobserved attributes that may have attracted more establishments to a given city (Henderson 2003).

Evidence suggests that the large estimated benefits are not a reflection of exogenous shocks or reverse causality. It could have been the case that some places may be intrinsically more productive, attract more workers, and thereby causing city size or population density to rise (“quantity” effects). Estimates that use instruments to control for endogeneity due to “quantity” effects do not drastically change the elasticity magnitudes. For example, in the U.S., Brazil, China, India and Brazil, the elasticity estimates remain stable when instrumenting current agglomeration with historical values. This dispels the fear that correlation between city size and productivity is caused by, for instance, in China by the post-1980 political shocks to particular areas, like the special economic zones (Chauvin et al., 2017). Likewise, Duranton (2016) does not find significant differences in elasticity estimates in Colombia with an IV estimation that use past population density or geological variables as an instrument.

2.4. Endogeneity – sorting and returns to skills

Agglomeration elasticity estimates may also be biased due to sorting of more educated workers or productive firms to large locations (“quality” effect). Firms endogenously choose locations based on a trade-off between production externalities and costs of production (e.g.,

¹ Otherwise for a gross-output production function another term reflecting intermediate inputs should be included.

² The validity of the instruments rely on the past populations being uncorrelated with unobserved drivers of contemporaneous wages conditional on the other controls. While a good case can be made for this (Combes et al., 2010), it is by no means decisive.

input costs). The productivity of a firm $\varphi(z, L, I_j)$ increases with its own 'raw' efficiency z and with local agglomeration externalities that depend on city size L . Yet, using French firm-level data [Combes et al. \(2012\)](#) find that more efficient firms disproportionately sort into more productive and larger cities, pointing to a potential endogeneity of firm productivity with city (size) can offer. In addition, more efficient firms can better leverage the local information they obtain and benefit from knowledge spillovers.

Likewise, workers' choice is also determined by similar trade-offs between wages and congestion costs. From equation (2), local average nominal wages can be shown as a function of labor skills and a composite local productivity effect. The latter is higher in larger cities because of presence of local (public) goods such as universities and opportunities for knowledge spillovers and dynamic learning ([De La Roca and Puga, 2017](#)).

To address sorting concerns one approach has been to include additional controls for skills, ideally at the worker or firm level, rather than the locality. While education and other observable characteristics of workers and firms can be controlled, unobservable traits that affect productivity may differ systematically across cities. A second approach is the inclusion of worker fixed effects when relating individual earnings to density ([Glaeser and Maré \(2001\)](#) and [Combes et al. \(2008\)](#)). The productivity benefits of density for workers is then identified from the changes in earnings that a given worker experiences when changing work location. Plant relocations are much less frequent than worker relocations and hence firm sorting is more difficult to control for.

Several approaches have been used to measure returns to skills. [Chauvin et al. \(2017\)](#) estimate elasticity of real wages as returns to human capital and find that the elasticity of density is lower for the US and China suggesting that the nominal-wage premium in dense locations do not reflect sorting of higher-ability people. Other studies have used observable skills and parental education and worker/individual fixed effects. Studies on the US find lower estimates of elasticity once controlling for skills (e.g. [Glaeser and Maré, 2001](#)). In Colombia, the elasticity of wages with respect to city size drops from 11 percent to 5.4 percent when individual worker characteristics such as education are included, implying that about half this relationship is explained by larger cities hosting more educated workers. This is consistent with the greater representation of more educated workers in larger cities and the view that workforce composition effects account for a sizeable fraction of spatial wage disparities ([Combes and Gobillon, 2015](#)).

In France, there is a high correlation between density and worker characteristics (0.44), and the associated elasticity of wage premium is cut by about one half when worker fixed-effects are included ([Combes et al., 2008, 2010](#)). Worker sorting is slightly weaker for Italy, where the correlation between individual fixed effects and density is 0.21 ([Mion and Naticchioni, 2009](#)). There is also evidence of spatial sorting in Spain and in the United Kingdom ([De La Roca and Puga, 2017](#); [D'Costa and Overman, 2014](#)).³

2.4.1. Manufacturing versus services

The productivity advantages of cities may differ by sector, as reflected by sector-specific parameter I_j in equation (1). In general agglomeration benefits for services are higher because firms rely more on face-to-face contact and are more likely to co-locate at a finer spatial scale (e.g. zip-code level), while manufacturing industries that trade with each other are more likely to co-locate in the same county or state. Moreover, agglomeration economies in services decay more rapidly with distance, making it more relevant for firms to cluster. For example, evidence from the advertising services industry supports an extremely rapid spatial decay of agglomeration effects that are shown to occur

primarily within 500 m ([Arzaghi and Henderson, 2008](#)). These results are broadly consistent with the findings of [Graham et al. \(2010\)](#) for the UK who find that the decay gradient is higher for services relative to manufacturing. Business services and consumer services have a decay gradient of 1.75 and 1.82 respectively, whereas for manufacturing the value is 1.10. Likewise, [Hasan et al., 2017](#) find that agglomeration effects in India are stronger for services relative to manufacturing.

More broadly, the difference in elasticity estimates between developed and developing economies are smaller when services sector is considered ([Fig. 2a](#)). In fact, they are generally based on individual earnings data that compensate for the cost of urban dis-amenities in developing world cities rather than total factor productivity (TFP). The estimated gap is smaller when comparing TFP estimates ([Fig. 2b](#)).

2.4.1.1. Spatial scale. Studies evaluate the spatial extent of local spillovers ranging from the broadest administrative level (regions) to the finest administrative level (e.g. villages and neighborhood). Most studies end up somewhere in the middle – administrative level 2 or 3 (e.g. municipalities or districts). The spatial scope of agglomeration effects depends on the nature of activity. For example, while knowledge and technology intensive activities will need to be co-located, other interactions such as input–output linkages can take place at a larger scale. A common approach is to consider an individual or location defined at a fine scale and to draw rings with increasing radius around it. Agglomeration benefits dissipate with distance and are rarely significant beyond a threshold distance and hence broader spatial scales yield lower benefits (see [Rosenthal and Strange, 2003](#); [Desmet and Fafchamps, 2006](#) for evidence on the US and [Rice et al., 2006](#) for the UK; [DiAddario and Patacchini, 2008](#) for Italy).

2.4.1.2. Underlying data. Elasticity estimates vary by the underlying data used for analysis. Spatially aggregated industry- or worker-level data might yield different elasticity estimates relative to micro firm- or worker-level data because with spatial aggregation the information on industry, firm and worker attributes is lost. Even within micro data, productivity measures derived from firm level data can include TFP, wages or labor productivity whereas those collected through household or labor force surveys typically reflect only wages. This may also be due to the type of firms in the data set. For instance, some countries have a cut-off on employee size in their firm census data, while comparable wage elasticity using household surveys may have a different threshold or criterion. In particular, elasticity estimates using firm level data might be lower than worker level data. For instance, [Tao et al. \(2019\)](#) find the wage elasticity for China to be negative 0.113 using firm level data compared to other studies that find huge estimates of ranging from 0.07 to 0.29 when using worker-level data ([Combes et al., 2019](#)).

2.4.1.3. Urban costs. Cities can predominantly be viewed as an outcome of a fundamental trade-off between agglomeration economies and urban costs ([Henderson, 1974](#); [Fujita and Ogawa, 1982](#)). As both firms and workers show tendencies for sorting into larger and productive locations, it increases dispersion forces in the form of urban costs. For example, the cost of immobile inputs such as land is higher in larger cities. The literature suggests that capturing true agglomeration benefits would require adjusting for pure externalities arising from local spillovers and dispersion forces pertaining to urban costs (e.g., congestion or housing/land prices). As firm productivity, wages and urban costs are simultaneously determined in equilibrium, including such costs as control variables in the estimation of agglomeration premia (e.g. land or housing prices) can lead to serious endogeneity biases. Alternatively, firm-level physical total factor productivity can also (partially) account for such costs when estimating net agglomeration benefits ([Grover et al., 2022](#)).

While much is known about the magnitude of the benefits of agglomeration economies, evidence about urban costs (the counterpart

³ For more evidence see [Bacolod et al., \(2009\)](#); [Abel, 2012](#); [Lindley and Machin \(2014\)](#) for the United States; [DiAddario and Patacchini, 2008](#) for Italy; [Groot and de Groot, 2020](#) for the Netherlands.

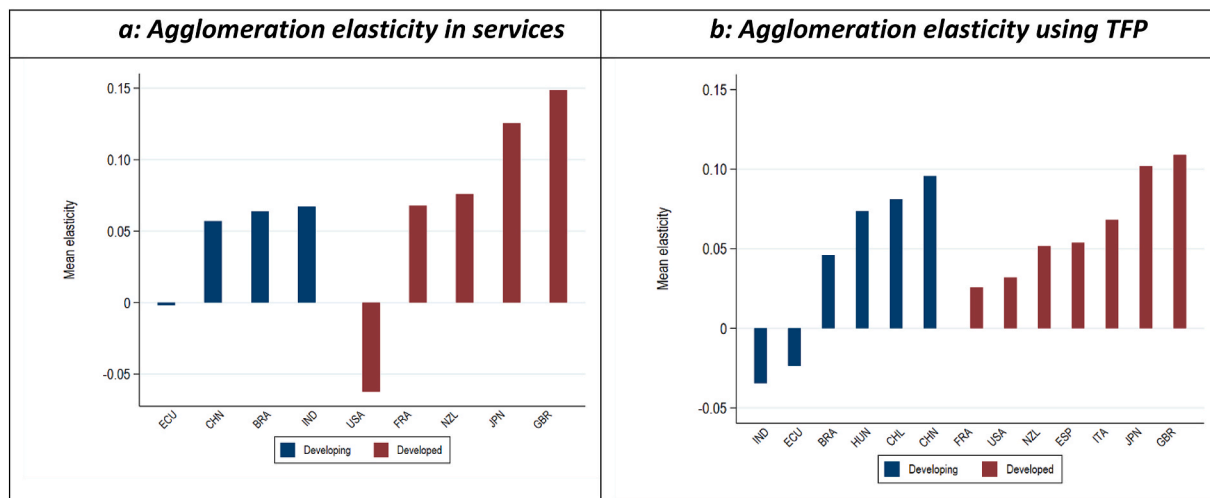


Fig. 2. Notes: Panel a computes unweighted average productivity elasticity estimates for each country using services sector data –encompassing both micro and spatial data, and wages, labor productivity and TFP estimates. Developed reflects studies of high-income countries, developing reflects non-high income, with income level determined using the World Bank country-income classification at the mid-year of each study's sample period. This reflects 193 raw elasticity estimates (58 in developing countries). Panel b computes unweighted average TFP elasticity estimates for each country – encompassing both micro and spatial data and any industry. This is derived from 470 estimates (64 in developing countries).

of agglomeration economies) is limited. Urban costs take a variety of forms. In larger cities, housing is more expensive, commutes are longer, and the bundle of amenities that these cities offer may differ. Residential mobility implies that urban (dis)amenities and commuting costs are reflected into land prices. The elasticity of urban costs with respect to city population is the product of three quantities: the elasticity of unit land prices at the city center with respect to population, the share of land in housing, and the share of housing in consumption expenditure. Urban cost has been proxied through price of consumption goods, commuting times and rents to income ratio and housing and land prices (Henderson, 2002; Combes et al., 2019; Baum-Snow and Pavan, 2011).

Although agglomeration benefits are higher, urban costs may explain why cities don't expand further. Urban costs have been estimated for France and Colombia using land and housing price data (Combes et al., 2013, 2019; Duranton, 2016). In France, the elasticity of urban costs with respect to population ranges from 0.016 to 0.05, while the estimates for agglomeration effects in France range from 0.015 to 0.03 (Combes et al., 2010). Similar results on the elasticity of urban costs are obtained for Colombia as well (Duranton, 2016). In both these countries, cities operate at near aggregate constant returns to scale suggesting that the benefits of agglomeration are balanced by their costs.

The costs of density on outcomes such crime and congestion is significantly high, and often higher than productivity elasticities. Ahlfeldt and Pietrostefani (2019) suggest that a log-point increase in density is associated with higher rents (0.15), mortality risk (0.09) and crime (0.085). To understand the true advantages of larger cities, it is critical to estimate agglomeration costs alongside benefits.

The overall effect of density on pollution depends on two opposing effects (Carozzi and Roth, 2023). On the one hand, dense locations have lower levels of vehicles miles traveled and hence lower pollution, while on the other hand the physical effect of density on concentration can offset these reductions in per capita emissions. Evidence from 1200 cities across the world suggests that denser cities exhibit lower emissions per capita and that pollution at the city level is characterized by the Environmental Kuznets Curve (EKC). Thus, income-emissions relationship follows an inverted-U pattern, with emissions per capita going up at early stages of development, but then declining as development proceeds (Castells-Quintana et al., 2021).

Long and unreliable travel times reduce the agglomeration benefits of large cities, whether for accessing jobs, markets, services or amenities. Travel is generally slow in developing countries even outside peak hours. For instance, Akbar et al. (forthcoming) show that the slowest large US city (New York) is generally faster than the fastest Indian city (Thrissur). Their work on cross-city comparisons in India concludes that denser, more populated cities are slower, and that there is a hill-shaped relationship between city per capita income and speed.

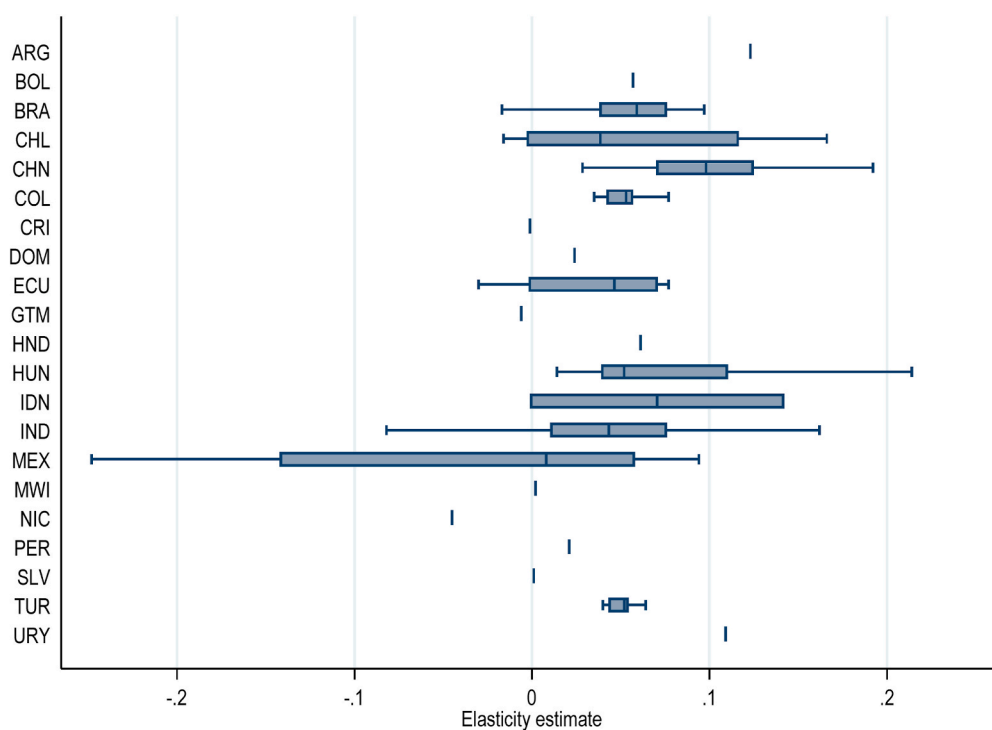
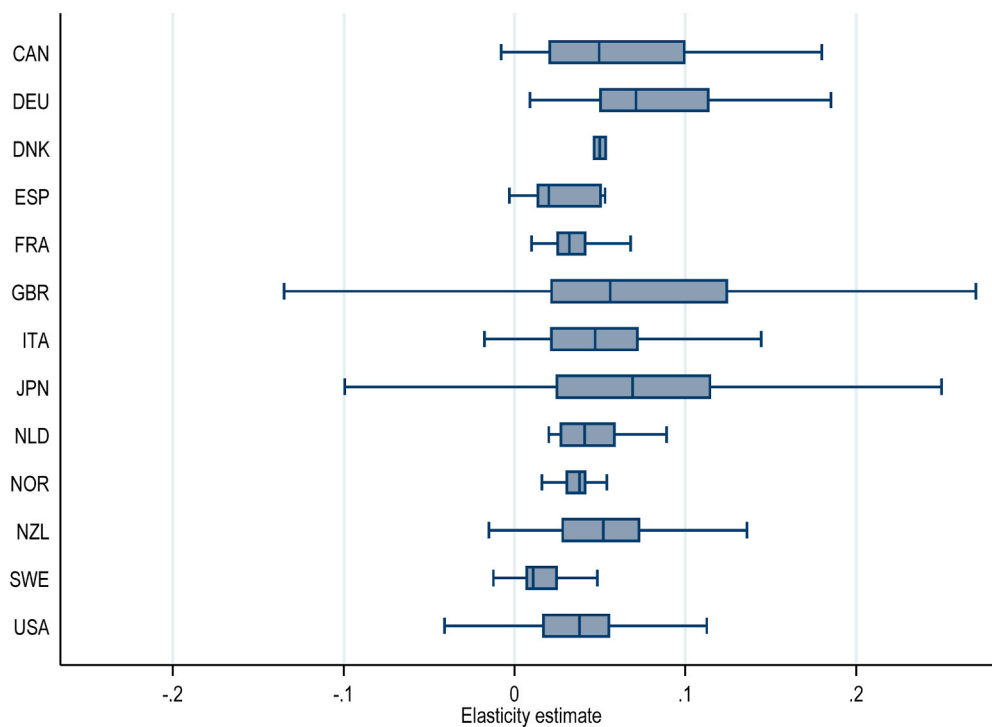
2.5. Urban costs and agglomeration economies – developing vs developed economies

There are several reasons why agglomeration economies or urban cost elasticities may differ across country development levels. Developing countries differ from that of advanced economies in a number of ways that affects both the productivity of their urban systems and the costs of city living.

First, labor markets in developing countries are typically characterized by duality, that is, a large informal sector co-exists with the formal sector. Low productivity in urban informal non-tradables hinders urban development (Jedwab et al., 2022). It also curbs the incentives for workers to improve their skills locally and thus limit the scope of agglomeration benefits (Duranton, 2015).

Second, land markets in developing cities are analogously characterized by duality in land use such that appropriate property titles and leases co-exist with squatted land, primarily due to the lack of effective, formal property titles (De Soto, 2000). Informal land markets coupled with outdated land regulations could breed misallocation of resources and may thus be a major barrier to enterprise development (Duranton et al., 2015).

Third, poor quality physical and transport infrastructure in developing countries increases urban costs and further reduces the scope of net agglomeration economies. Evidence from India (Ghani et al., 2016), Ghana (Jedwab and Moradi, 2012), China (Baum-Snow et al., 2017), and Africa (Storeygard, 2016) underscore the importance of market access and the profound and long-lasting effects of transportation infrastructure. Cities in developing countries face the same downsides of density, that is, congestion, crime and diseases, however, they lack the

Panel A – Developing Countries**Panel B – Developed Countries****Fig. 3.** Elasticity estimates by Country

Notes: The box shows the 25th, median and 75th percentile elasticity for each country, with the whiskers showing the upper and lower adjacent values. Some countries reflect a single estimate, hence only the median is shown.

Table 1
Summary statistics of controls included in meta-analysis.

Category	Variable	Details	Reference Category	mean	std dev
Country Income	Developing Country	= 1 if estimates reflect a non-high-income country, using World Bank classification at mid-point of study period	estimates reflect high-income country	0.294	0.456
Productivity Measure	Wages	= 1 if estimates reflect wage or earnings outcome	estimates reflect TFP outcomes (either TFP specifically, or outcome is an output measure in combination production function estimation)	0.468	0.499
	Labor Productivity	= 1 if estimates reflect labor productivity outcomes (output or value-added per worker)		0.165	0.372
Urban Cost	Accounts for Urban Costs	= 1 if estimations include a local measure of urban costs, e.g. land rents, house prices, local price indices	estimations do not include a measure of urban costs	0.031	0.174
Industry	Manufacturing Sector	= 1 if estimates reflect the manufacturing sector	estimates are not specific to manufacturing or services	0.390	0.488
	Services Sector	= 1 if estimates reflect the services sector		0.151	0.358
Skill	Skilled workers	= 1 if estimates reflect skilled workers (including managers, scientists or non-routine occupations)	estimates are not specific to skilled or unskilled workers	0.020	0.141
	Unskilled workers	= 1 if estimates reflect unskilled workers (routine occupations)		0.026	0.158
Time Period	Post-1990	= 1 if mid-point of data is 1990 onwards	mid-point of data is before 1990	0.788	0.409
Study Quality	Published	= 1 if study is published in a peer-reviewed journal		0.736	0.441
	Number of Citations (as of April 2020)	Log number of citations - normalized by study publication year	–	0.000	1.357
Spatial Measure	City - Level	= 1 if study uses city-level spatial units	study uses regional level spatial units	0.537	0.499
	Sub City - Level	= 1 if study uses sub-city-level spatial units		0.304	0.460
Agglomeration Measure	Density Measure	= 1 if agglomeration is measured by population or employment density	agglomeration is measured by size, such as total employment or population	0.450	0.498
	Market Potential Measure	= 1 if agglomeration is measured by effective density or market potential		0.249	0.433
Localization	Localization Control	= 1 if a localization control is included (reflecting the size of own industry)	estimations do not include a localization control	0.303	0.460
Data	Panel Data	= 1 if study uses panel data	study uses cross-section data	0.595	0.491
	Firm Data	= 1 if study uses firm-level micro data	study does not use micro data	0.341	0.474
	Worker Data	= 1 if study uses worker-level micro data		0.407	0.491
Endogeneity	Panel Fixed Effects	= 1 if panel fixed effects are included or estimation is in differences	study does not use fixed effects	0.210	0.407
	IV estimation - contemporaneous	= 1 if estimates reflect IV estimation, using contemporaneous or short-lagged instruments (including GMM)	study does not use IV estimation	0.067	0.250
	IV estimation - historic	= 1 if estimates reflect IV estimation, using historic or geological instruments		0.169	0.375
Firm Heterogeneity	Industry control	= 1 if estimates include an industry dummy variable	study does include industry or firm size controls	0.257	0.437
	Firm size control	= 1 if estimates include a firm size control		0.372	0.483
Worker Sorting	Local area human capital	= 1 if study includes a local area human capital measure	study does not control for human capital	0.183	0.387
	Individual-level human capital	= 1 if study includes worker-level controls		0.276	0.447

Note: Number of observations for each variable is equal to 1281. We normalize patents by publication year, common practice in the patent literature, by using the residual from a regression of citations on publication year.⁵¹ We reported unweighted means and standard errors. Country-income is captured by a dummy variable reflecting whether the country is non-high-income country, using the mid-point of the study's data sample.⁶²

Table 2
Summary statistics of variables included in urban cost elasticity analysis.

	Developing			Developed		
	n	mean	std dev	n	mean	std dev
PM2.5 Pollution	76	3.289	0.592	275	2.659	0.502
CO2 Emissions	120	0.316	0.978	69	1.334	0.980
Hours Lost to Travel Time	73	4.940	0.332	280	4.606	0.320
Wellbeing	106	1.640	0.159	59	1.887	0.100
Homicide Rate	634	1.923	1.369	469	0.463	0.908
Population Density in 2015	634	9.551	0.522	469	8.819	0.401
Night Lights	634	4.582	0.977	469	5.244	0.715
Population Density in 1975	634	9.213	0.603	469	8.616	0.839
Sub-soil Clay Percentage	634	28.046	16.259	469	18.481	15.122
Sub-soil Sand Percentage	634	37.620	17.335	469	32.636	19.320
Sub-soil Silt Percentage	634	26.527	11.741	469	19.998	13.396

Note: All variables are in logs, with the exception of the sub-soil percentages. All variables reflect city-level data.

financial resources to invest in infrastructure such as transport network, clean water and air, and sewerage to mitigate these risks. Given the costly investments that they require, developing country cities typically remain disconnected (Lall et al., 2017).

Lastly, developing country cities are also more likely to experience urban favoritism by governments toward the largest cities (Ades and Glaeser, 1995; World Bank, 2020). This harms both the favored primate city by making it more crowded and congested and other cities which are, in effect, heavily taxed (Duranton, 2008). Finally, developing countries lack the government capacity to enforce urban regulations (Glaeser and Poterba, 2020).

3. Data

We rely on two separate datasets: one for the meta-analysis of agglomeration elasticities, and another for the analysis of urban costs. First, we discuss the construction of the sample of papers used for the meta-analysis, and second, the urban cost data.

3.1. Meta-analysis data

The primary focus of our paper is a meta-analysis of agglomeration productivity elasticities differentiating between developing and developed countries, where our paper follows the approach outlined in Stanley et al. (2013) to construct the sample of studies (further details are given in the Appendix).

We build on the meta-analysis data of Melo et al. (2009) and Ahlfeldt and Pietrostefani (2019) by extending their sample on elasticity estimates for developing countries. We use a combination of keywords, such as “agglomeration” or “density” or “urban”, and “elasticity” or “productivity” or “wages”, applied to the academic databases EconLit, Web of Science, and Google Scholar. We supplement the keyword search by conducting an analysis of citation trees of key published and working papers by eminent researchers in the field and papers that inspired the study.⁴ We also consulted with experts in academia (including the University of Pennsylvania, John Hopkins University, and Oxford) and practice (including the World Bank) to identify ongoing work in the field.

⁵ An alternative approach of demeaning citations using the mean citations of publications in that year is not possible, due to few observations in some publication years.

⁶ The analysis is robust to defining country income using the first or last period of the study sample.

⁴ Melo et al. (2009), Chauvin et al. (2017) and Ahlfeldt and Pietrostefani (2019).

The initial sample included peer-reviewed papers, working papers of universities, research institutes and international organizations (World Bank, NBER, CEPR, CESifo, and IZA), chapters in books and conference proceedings.

We restrict the initial sample to papers with one of four outcome variables — labor productivity, wages or earnings, TFP and output using a production function – and four agglomeration measures: population size, population density, economic density and market potential. To ensure comparability, we only retain studies that either estimate unit-free agglomeration elasticities, or those that could be converted to an elasticity (e.g. semi-elasticities). Estimates reporting analysis pooled across several countries are excluded; country specific estimates are retained.

To test and correct for publication bias, we also collect standard errors of estimates either from the papers or by requesting the authors. Studies without the standard errors needed for precision-weighting are excluded from the meta-analysis. Most studies are empirical; however, studies estimating structural models are also included provided they report agglomeration elasticities and standard errors. To avoid double counting several versions of the same study, in particular for working papers, we include only the latest version.

Our resulting sample comprises of 1281 elasticity estimates sourced from 76 studies across 34 countries covering 1973 to 2020. Of the 76 studies, 29 are reported in Melo et al. (2009), 16 are drawn from Ahlfeldt and Pietrostefani (2019), with 29 additional studies included here. Table A8 in the Appendix reports the full list of 76 studies. Three hundred and seventy-six estimates are for developing countries.

Even within a single study or a single country in our data there are often a wide range of estimates, highlighting substantial heterogeneity in data and estimation methods. Fig. 3 reports a forest plot of the estimated 1281 agglomeration elasticities across the 34 countries – with estimates from developing countries presented in Panel A and developed countries in Panel B. The figure suggests that countries show a broad range of elasticities, with some reporting both large positive and negative estimates. While some of the variation is likely a result of differences in time period, spatial scale, estimation methods and so on across papers, Appendix Figure A1 shows a similar spread in estimates within individual studies. The wide range of estimates strongly suggests key differences in underlying factors within and across studies, highlighting the importance of controlling for such distinctions through a meta-analysis.

Table 1 summarizes the variables collected for each study that act as control variables in the meta-analysis, while Appendix Table A1 summarizes the estimated agglomeration elasticities by each meta control category. In Appendix Table A2 we further summarize the control variables across developed and developing country studies. Out of the 376 developing country estimates, whilst only 45 include panel fixed effects to address endogeneity 142 include historic or geological IVs and 237 include worker-level human capital controls to tackle worker sorting. By comparison, our analysis includes 905 developed country estimates, where 225 include panel fixed effects and only 75 include historic or geological IVs and 116 include worker human capital controls. The number of estimates employing higher quality estimation approaches that address endogeneity or worker sorting is broadly similar in the two income groups.

3.1.1. Urban cost data

We also provide novel estimates of urban cost elasticities across developing and developed countries. While Ahlfeldt and Pietrostefani (2019) provide a good synthesis of the available literature, they focus on developed countries. We use five measures of city-level urban costs: congestion (hours lost due to traffic), pollution (PM2.5 emissions, CO2 emissions), subjective wellbeing and crime (homicide rate).

We use two measures of pollution. PM2.5 emission levels are sourced from the World Health Organization and is a cross-section of data with information available for 2013 or 2014, for a sample of 351 cities (with

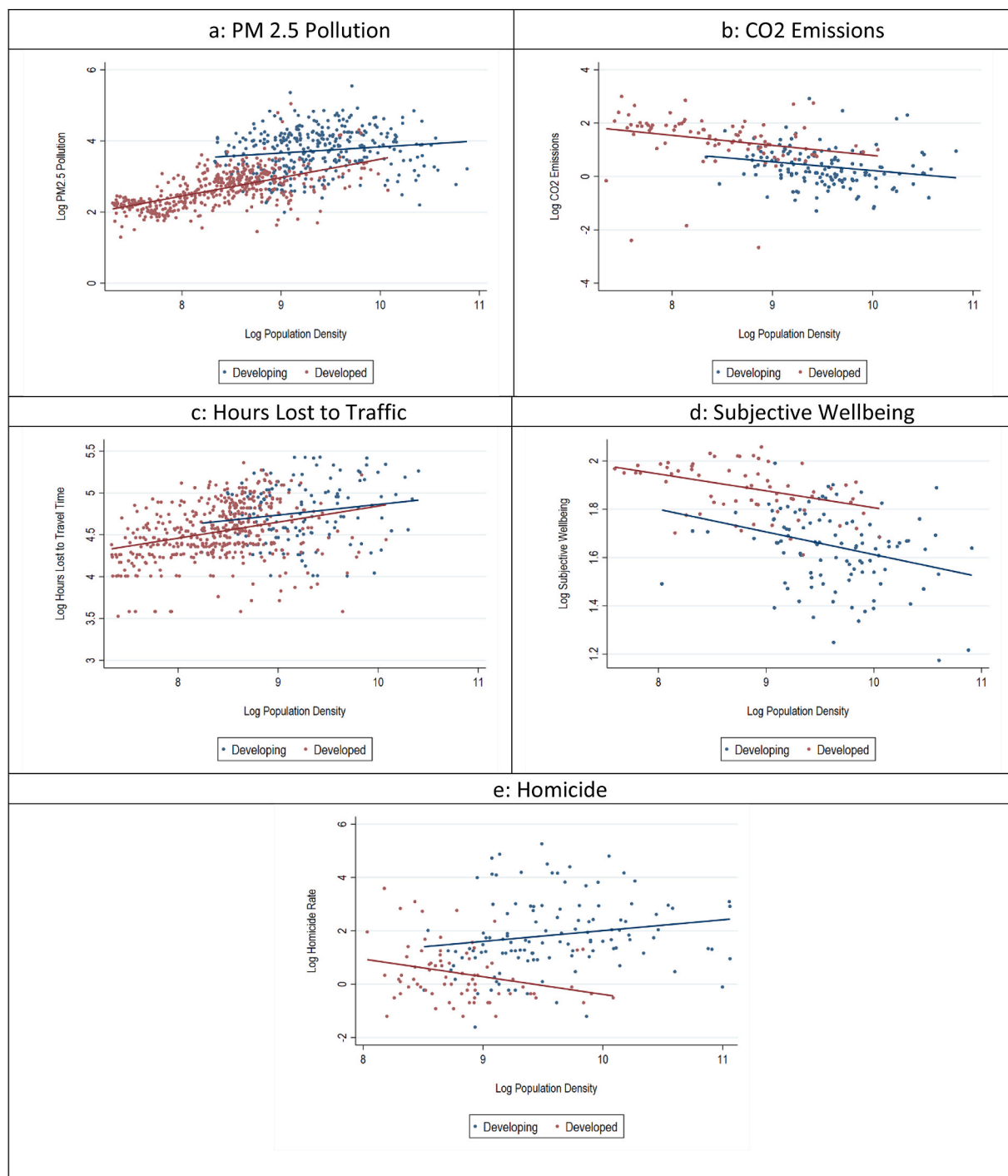


Fig. 4. Urban costs with city density in developed and developing countries.

Note: Population density measured in 2015. Homicide data reflects the first available year of city data, all other information is a cross-section as noted above. Best fit lines are superimposed.

Table 3
Funnel asymmetry test for publication bias.

Estimation Method:	(1) Meta RE	(2) Meta FE	(3) WLS
Standard error (publication bias)	1.196*** (0.087)	3.810*** (0.031)	3.810*** (0.574)
Constant	0.039*** (0.002)	0.012*** (0.000)	0.012*** (0.004)
Observations	1281	1281	1281

Notes: Estimation of equation (4). Meta FE denotes meta-analysis fixed effects estimation, Meta RE reflects meta-analysis random effects estimation, WLS are precision-weighted least squares estimates, using the inverse of the reported estimate's variances as weights. Outcome is the reported agglomeration elasticity expressed as absolute values. We do not exclude outliers from our baseline analysis, but rather include an extensive set of controls to explain study heterogeneity.¹²¹ Robust standard errors are presented in parentheses and clustered at the study level for OLS and WLS, clustering is not possible under the Meta RE and Meta FE commands. ***, ** and * reflect significance at the 1, 5 and 10 percent levels respectively.

76 in developing countries). We exclude China from the pollution analysis because of potential data reporting concerns highlighted in Greenstone et al. (2021). CO2 emissions per capita are taken from Nangini et al. (2019) and is a cross-section of data with data available of 2010–2015 for 189 cities (120 in developing countries). We include two measures of congestion. Hours lost to travel time information is sourced from TomTom, which provides real time traffic statistics for over 600 million users around the world. The measure reflects the average annual additional hours spent driving in rush hours in 2018 and is available for 353 cities, of which 73 cities are in developing countries. Subjective wellbeing (SWB) score measures the overall valuation of life on a scale of 0–10. An average value of SWB for 2014–2018 is an average value for 2014–2018 available from De Neve and Krekel (2020) for 165 cities (106 in developing countries). Crime is measured using the homicide rate obtained from United Nations Office on Drugs and Crime (UNODC) global study on homicide (2019) and is available from 2003 to 2017. We utilize an unbalanced panel of crime data for 2006 to 2016, comprising 234 cities (with 43 in developing countries).

City-level population density is calculated as the population per square kilometer of *built-up density* using 2015 data from the Global Human Settlement Layer (GHS). Unfortunately, this data is only available at irregular intervals (1975, 1990, 2000 and 2015), which combined with limited time horizon of our urban cost measures, limits us to a cross-section analysis. We also examine robustness to population density from country census data that are available for approximately half the cities in our baseline sample.

Average city-level night lights is calculated from the Global Radiance Calibrated Nighttime Lights database from the National Centers for Environmental Information. City-level geological variables reflecting the average sub-soil composition are calculated from the Harmonized World Soil Database. We use geospatial data on city boundaries for the nearly 700 cities in developed and developing countries for which we have an urban cost measure – using boundary data from GADM. GADM contains data at various levels of spatial aggregation that are not harmonized across countries. We individually inspected the spatial aggregation levels for every city and country and compared to Google Maps to choose the GADM level which most closely matched the city boundary in Google Maps. For each city we then report the average night lights, or geological variables.

Summary statistics of these variables are provided in Table 2. Fig. 4 shows how pollution, crime, wellbeing and congestion vary by city density in both developed and developing countries, giving a suggestion of our later regression analysis. For most series we see higher levels of urban costs in developing country cities for any given level of density: PM2.5 pollution, hours lost to traffic and homicides are higher and subjective well-being is lower. However, CO2 emissions appear to be higher in developed economies. In terms of the relationship with

density, these appear to be similar in developed and developing countries, with the exceptions of homicides. We examine this in more formal regression analysis later.

4. Estimation

We first discuss the estimation strategy for the meta-analysis of agglomeration elasticities, followed by the methodology for estimating urban cost elasticities.

4.1. Meta-analysis estimation

There are several estimation challenges in conducting a meta-analysis (see Florax, 2001; Stanley, 2008).

First, selective reporting can lead to bias in meta-analyses. Publication bias can reflect a preference among researchers and journals to selectively report statistically significant coefficients in a desirable direction, for instance, those in agreement with established theory. Such selective reporting implies that the reported estimates are truncated, that is, we primarily observe estimates that are statistically significant. This introduces a systematic relationship between reported estimate e_i for study i and the reported standard errors, se_i . More formally, as Stanley and Doucouliagos (2014) note, the first-order approximation of the conditional expectation of a truncated normal distribution implies the reported estimates will be a linear function of the estimates' standard errors. We follow standard practice and test for publication bias using the funnel asymmetry test (FAT) (e.g. Card and Kreuger, 1995; Egger et al., 1997; Stanley, 2008):

$$e_i = e_0 + \beta_0 se_i + u_i \quad (4)$$

The null hypothesis of no publication bias corresponds to $\beta_0 = 0$, i.e. no relationship between the reported study effects and the reported standard errors. We use the absolute value of elasticity estimates in estimation (1) such that the alternative hypothesis is selective reporting of positive or negative statistically significant results. Since more than 90 per cent of the elasticities we use are positive, in practice this makes little difference. To test for publication bias, equation (4) can be estimated using a variety of models: unweighted OLS, meta-random effects, meta-fixed effects and precision-weighted least squares (WLS).⁷

Traditional methods can be particularly sensitive to publication bias (Stanley and Doucouliagos, 2014, 2019). We deploy state of the art estimators that outperform these approaches. More specifically,

⁷ Note that we use the terms meta-random effects and meta-fixed effects to reflect “random effects” and “fixed effects” meta analyses, in order to distinguish them from the similarly named (but distinct) panel estimators.

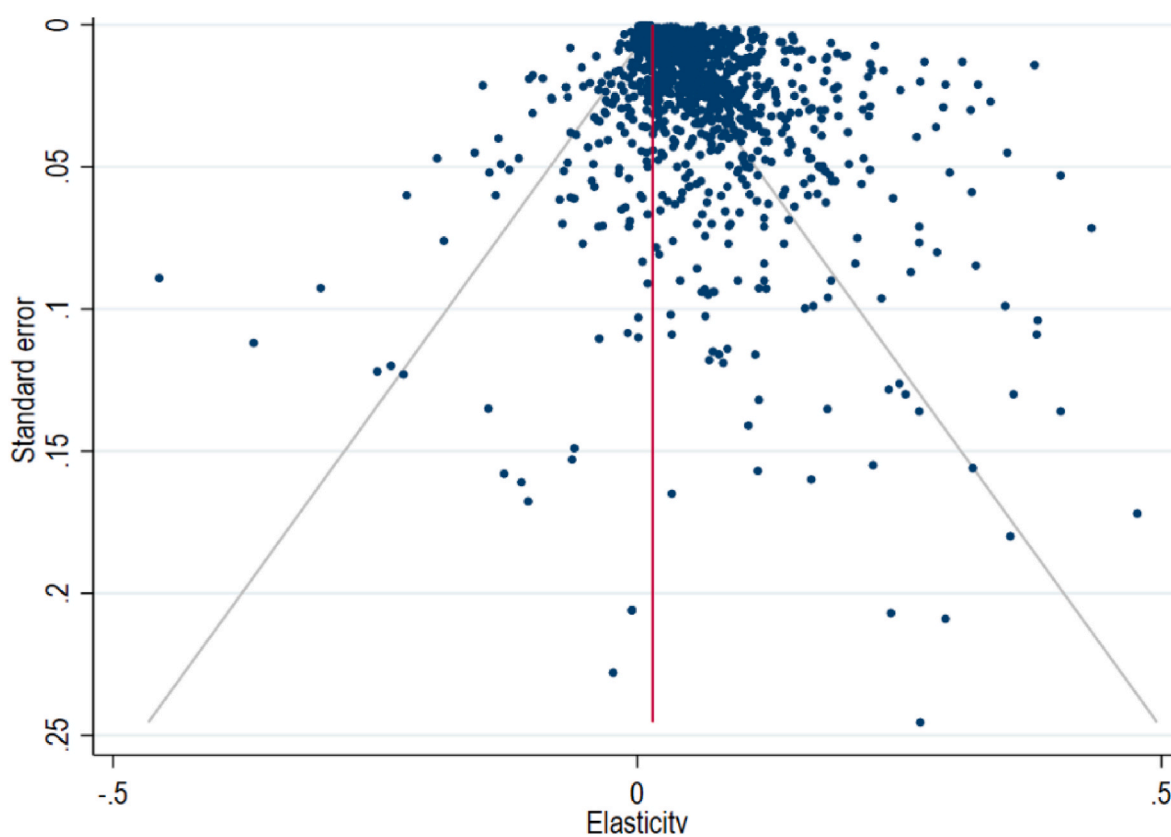


Fig. 5. Funnel Asymmetry Test Plot of Agglomeration Elasticity Estimates against Standard Errors

Notes: Elasticity estimates are plotted against standard errors. The red line represents the constant and grey funnels represent 95 percent confidence intervals of a t -test of publication bias, from a meta-random effects estimation (as in column 1 of Table 3). In the absence of publication bias, one would expect 95 percent of elasticity estimates to lie within the funnel and to be evenly distributed. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

Table 4
Funnel asymmetry test for heterogeneity in publication bias across study characteristics.

	(1)	(2)	(3)	(4)	(5)	(6)
Estimation Method:	WLS					
Standard error	4.430*** (0.713)	−0.349 (1.065)	3.538*** (1.049)	4.066*** (0.545)	1.968*** (0.520)	0.187 (1.608)
* Developing Country	−2.573*** (0.879)					−2.859*** (0.949)
* Post-1990		5.027*** (1.213)				5.919*** (1.599)
* Published			−0.442 (1.127)			0.761 (1.168)
* Number of Citations				−0.884 (0.671)		−0.477 (0.358)
* Micro Data					2.621*** (0.842)	−2.065 (1.877)
Constant	0.011*** (0.004)	0.038*** (0.014)	0.029*** (0.008)	0.007** (0.003)	0.010* (0.006)	0.040** (0.017)
Observations	1281	1281	1281	1281	1281	1281

Notes: Outcome is the reported agglomeration elasticity. WLS is precision-weighted least squares estimates, using the inverse of the reported estimate's variances as weights. WLS estimation of equation (4), but interacting the estimates' standard errors with different study characteristics to allow for heterogeneous publication bias across studies. Non-interacted study characteristics are included in the estimation, but not reported for parsimony. Study characteristics are defined in Table 1. Reported agglomeration elasticity expressed as absolute values. Robust standard errors clustered at the study level are in parentheses. ***, ** and * reflect significance at the 1, 5 and 10 percent levels respectively.

Precision-effect estimate with standard error (PET-PEESE) has been shown to outperform several estimators, such as unrestricted WLS or WAAP (Weighted Average of Adequately Powered studies) or random effects estimation under publication bias (Stanley and Doucouliagos, 2015, 2019). Unrestricted WLS or random effects do not account for publication bias, although unrestricted WLS, by incorporating precision weighting, has been shown to outperform random effects under such bias (Stanley and Doucouliagos, 2014). WAAP passively reduces publication bias since it excludes low-powered estimates and low-power has been found to be correlated with publication bias (Stanley and Doucouliagos, 2019). In contrast, PET-PEESE incorporates precision weighting and also proxies publication bias using a quadratic approximation to the conditional expectation of a truncated normal distribution, thus assuming the underlying estimates are normally distributed with a truncation due to selective reporting. So, PET-PEESE includes the square of the estimates' standard errors, in place of a linear relationship as in equation (4).⁸ All the estimation methods presented are typically precision-weighted, meaning that they weighted by the inverse of the estimates' variances (se_i^2), such that more precise estimates are more heavily weighted than more noisy estimates. We use PET-PEESE as our preferred specification.

Our baseline specification, PET-PEESE, involves estimating the following precision-weighted regression:

$$\hat{e}_i = e_0 + \alpha_1 ldc_i + \gamma Z + \beta_0 se_i^2 + \beta_1 se_i^2 ldc_i + u_i \quad (5)$$

where, ldc_i is a dummy variable for studies of developing countries and α_1 is our coefficient of interest and reflects differences in agglomeration elasticities across developing and developed countries. All regressions are weighted by the inverse of the estimates' variances (se_i^2), e_0 is the constant term, and in addition to the estimates' variance term to capture publication bias, we add interactions with ldc_i , to understand differences in the publication bias between countries in the two income groups.⁹ Z is

a vector of study characteristics to allow for heterogeneity across studies, outlined in Table 1. All estimations include cluster-robust standard errors, clustered within studies, to allow for correlation across estimates within a study and also mitigate potential type I errors (as suggested by Stanley and Doucouliagos, 2019).

Second, since reported estimates can vary by estimation methods, country, time period, unit of analysis, and degree of spatial aggregation among many other elements, the specification of included study characteristics can critically affect results. The vector of controls, Z are listed in Table 1. The broad set of potentially relevant study characteristics extends those included by the previous meta-analyses of Melo et al. (2009) and Ahlfeldt and Pietrostefani (2019) to encompass the recent literature as discussed in section 2. However, the choice of which sub-set of study controls to include as meta-regressors is not straight-forward.

We employ two techniques to strip-off researcher judgement and potential biases in selecting study controls. First, we run all 17 million combinations of covariates, with standard errors clustered at the study-level, and choose the parsimonious models that minimize the Bayesian Information Criteria or the Akaike Information Criteria. Both BIC and BMA are consistent, meaning that they converge to the true model, and BIC provides a good approximation of many linear models (van Erven et al., 2012; Fragoso et al., 2018). Cluster-robust estimation using BIC model selection is our preferred specification. Second, we also employ Bayesian Model Averaging (BMA) as a robustness to our preferred variable selection approach. BMA involves estimating millions of regressions consisting of subsets of the potential explanatory variables and weights them by model fit and model complexity (following Havranek et al., 2017; Steel, 2020). BMA reports an average of these many underlying regressions. The only covariates required to be included in each regression are the variance terms (to control for publication bias) and the non-interacted developing country dummy and post-1990 terms – we estimate nearly 17 million regressions for every combination of other covariates. We use BMA as a robustness, rather than our preferred specification, since BMA does not report clustered standard errors¹⁰.

Third, meta-analyses that take a single “preferred” estimate per study have the risk of overweighting smaller studies that contribute to only a

¹² Our results are robust to winsorizing the top and bottom 1percent of estimates by country.

⁸ Our results are also robust to including higher order standard error terms, for a cubic approximation of the truncated normal distribution. Results are available upon request.

⁹ In section 5a, we also add interactions between the estimates' variances and other data characteristics, to test for heterogeneous publication bias across data samples.

¹⁰ Since most econometric studies present multiple estimates, these estimates may not be statistically independent within a study. Clustered standard errors allow for such dependencies (De Luca and Magnus, 2011; Havranek et al., 2017).

Table 5

Homogeneous meta-estimates – agglomeration elasticities across developed and developing countries.

Estimation Method:	(1) Meta RE	(2) Meta FE	(3) WLS	(4) PET-PEESE	(5)
Developing Country	0.009** (0.004)	0.022*** (0.001)	0.022** (0.009)	0.021** (0.009)	0.021** (0.010)
Post-1990s					–0.025* (0.013)
Constant	0.050*** (0.002)	0.014*** (0.000)	0.014*** (0.004)	0.014*** (0.004)	0.037*** (0.013)
Observations	1281	1281	1281	1281	1281

Notes: Outcome is the reported agglomeration elasticity. Meta RE reflects meta-analysis random effects estimation, Meta FE reflects meta-analysis fixed effects estimation, WLS are precision weighted least squares estimates, using the inverse of the reported estimate's variance as the weight. PET-PEESE are "precision-effect estimates with standard errors" following [Stanley and Doucouliagos \(2019\)](#), which to correct for publication bias includes a squared standard error term in column 4, and its interaction with a developing country dummy and with a post-1990s dummy in column 5. The standard error terms are omitted for parsimony. PET-PEESE and WLS are precision-weighted, using the inverse of the reported estimate's variance as the weight. Robust standard errors are presented in parentheses, these are clustered at the study level for all except Meta RE and FE estimation. ***, ** and * reflect significance at the 1, 5 and 10 percent levels respectively.

small number of estimates (and underweighting larger studies that produce multiple estimates). This also raises complications in choosing a single "best" estimate from each study, particularly when even within a single study there can be a range of estimation methods or data sample restrictions employed (e.g. manufacturing vs services, worker skill). Selecting a single estimate also requires researcher judgement, which may inadvertently introduce bias in the meta-analysis.

Instead, for our focal analysis, we follow the most common practice and retain multiple estimates per study, and include an exhaustive set of potential covariates to explain heterogeneity across estimates and address issues of within-study autocorrelation by clustering at the paper-level. Our sample of 1281 elasticity estimates are drawn from 76 studies and range from a single estimate in some papers to up to 80 estimates in a given study. To ensure our results are not driven by a few studies with a large number of estimates, we do a robustness check by taking a single estimate per combination of meta controls. To avoid introducing researcher biases in choosing this estimate, the single estimate is the mean elasticity reported in the study.

4.2. Urban cost estimation

We estimate urban cost elasticities across developed and developing countries using city-level OLS regressions of (log) urban cost against (log) population density (equation (6)). We use five measures of city-level urban costs: congestion (hours lost due to traffic), pollution (PM2.5 emissions, CO2 emissions), subjective wellbeing and crime (homicide rate).

A developing country dummy is included to reflect difference in urban costs in levels, and interact the dummy with population density to estimate the difference in urban cost *elasticities* across income groups. We normalize density relative to the mean value, such that constant terms can be interpreted as the levels at the mean city density.

Specifically, we estimate:

$$cost_i = \delta_0 + \delta_1.ldc_i + \delta_2.density_i + \delta_3.ldc_i * density_i + \delta_4.Z_i + u_i \quad (6)$$

where $cost_i$ is the log urban cost for city i , urban costs reflect either the log average annual additional hours spent driving in rush hours, log PM2.5 emissions, CO2 emissions, subjective well-being or log homicide rates. ldc_i is a dummy variable reflecting cities in developing countries, $density_i$ is the log population density for city i in 2015. Developing countries reflects non-high-income status as defined in 2015, using the World Bank classification. Z_i is a vector of controls. We include a city-level control for average night lights in all specifications. The homi-

cide data is available for an unbalanced panel of cities, consequently we include year fixed effects as additional controls, estimating a pooled cross-section regression.¹¹

We suspect potential endogeneity concerns when estimating urban cost elasticities. This could be due to missing variables that are correlated with both population and urban costs; or a reverse causation that leads more expensive cities to end up being smaller. Omitted factors such as income or productivity correlate with density and also with industrialization that causes pollution or with car use is associated with congestion. Moreover, because the fundamental trade-off between agglomeration benefits and costs that shapes the evolution of density, it can be difficult to disentangle the causal effect on urban costs. One way to address this concern is to control extensively for the characteristics of cities to preclude these sources of correlation with the error term. We include a control for city-level night-lights in all specifications, which has been widely used a proxy for economic activity, one obvious omitted factor (e.g. [Henderson et al., 2018](#) amongst others). However, due to lack of comparable data on city characteristics across the globe we are unable to include other measures.

To further mitigate endogeneity concerns we present instrumental variable (IV) estimation. [Combes et al. \(2019\)](#) employ a variety of approaches to address endogeneity, including long lags of urban population and density, following the standard urban literature estimating agglomeration effects ([Ciccone and Hall, 1996](#); [Combes et al., 2008](#); [Combes and Gobillon, 2015](#)). Our primary IV analysis uses historic instruments – population density in 1975 – to predict density in 2015. Given that the development and urbanization of many developing countries was slow until recent decades, it seems likely that the historic density is a plausibly exogenous instrument, at least for developing countries. Research on single developed countries (like [Combes et al., 2019](#)) has often used older census data as instruments, unfortunately this is not possible to obtain across developing countries – even contemporaneous census data is available for less than half our sample of cities.

As a secondary IV analysis we construct novel geological instruments of the subsoil composition that predicts the suitability for tall buildings (e.g. following [Combes et al., 2019](#)). Our use of geological variables can be rationalized because such factors mattered in the past by making an urban area a reasonable place to live centuries ago differ from the factors that drive urban costs today. These instruments for city size would be

¹¹ Unfortunately there is too little time variation within cities to estimate a panel fixed effects specification and we have too few cities within most developing countries to include country fixed effects.

Table 6
Heterogeneous meta-estimates by study characteristics.

		(1)	(2)	(3)	(4)
		PET-PEESE			
Estimation Method: Model Selection		General	BMA	AIC	BIC
Country Income	Developing Country	0.005 (0.007)	0.004 (0.003)	0.004 (0.007)	0.005 (0.008)
Productivity Measure	Wages	0.064*** (0.013)	0.059*** (0.006)	0.064*** (0.014)	0.060*** (0.013)
	Labor Productivity	0.036*** (0.011)	0.036*** (0.004)	0.037*** (0.013)	0.036*** (0.013)
	Accounts for Urban Costs	−0.041*** (0.012)	−0.039*** (0.006)	−0.042*** (0.013)	−0.042*** (0.013)
Urban Cost	Accounts for Urban Costs	−0.041*** (0.012)	−0.039*** (0.006)	−0.042*** (0.013)	−0.042*** (0.013)
Industry	Manufacturing Sector	0.017** (0.007)	0.016*** (0.002)	0.017** (0.007)	0.016** (0.007)
	Services Sector	0.037*** (0.011)	0.037*** (0.004)	0.037*** (0.011)	0.037*** (0.011)
Skill	Skilled workers	0.013*** (0.004)	0.005 (0.008)	0.013*** (0.004)	
	Unskilled workers	−0.003*** (0.001)	−0.002 (0.002)	−0.003** (0.001)	−0.003** (0.001)
Time Period	Post-1990s	−0.001 (0.007)	−0.002 (0.003)	−0.001 (0.006)	−0.003 (0.006)
Study Quality	Published	−0.004 (0.005)	−0.004 (0.003)	−0.004 (0.005)	−0.005 (0.005)
	Number of Citations	−0.000 (0.002)	−0.000 (0.000)		
Spatial Measure	City - Level	0.002 (0.002)	0.000 (0.001)		
	Sub City - Level	−0.012** (0.005)	−0.012*** (0.002)	−0.013*** (0.003)	−0.013*** (0.003)
Agglomeration Measure	Density Measure	−0.001 (0.007)	−0.000 (0.001)		
	Market Potential Measure	−0.005 (0.006)	−0.007** (0.003)	−0.005 (0.007)	−0.006 (0.007)
Localization	Localization Control	0.005 (0.006)	0.002 (0.003)	0.005 (0.005)	0.003 (0.005)
Data	Panel Data	−0.004 (0.005)	−0.001 (0.002)	−0.005 (0.005)	
	Firm Data	0.047*** (0.012)	0.044*** (0.005)	0.047*** (0.011)	0.044*** (0.012)
	Worker Data	−0.017** (0.008)	−0.015*** (0.005)	−0.017* (0.009)	−0.016* (0.008)
Endogeneity	Panel Fixed Effects	−0.021*** (0.006)	−0.021*** (0.002)	−0.021*** (0.006)	−0.021*** (0.006)
	IV estimation - contemporaneous	−0.001 (0.005)	−0.000 (0.001)		
	IV estimation - historic	−0.001 (0.007)	−0.000 (0.000)		
Firm Heterogeneity	Industry control	0.011*** (0.004)	0.012*** (0.002)	0.011*** (0.003)	0.010*** (0.003)
	Firm size control	−0.004** (0.002)	−0.003** (0.001)	−0.004** (0.002)	−0.004* (0.002)
Worker Sorting	Local area human capital	−0.010** (0.004)	−0.009*** (0.003)	−0.010** (0.004)	−0.010** (0.004)
	Individual-level human capital	−0.014** (0.005)	−0.014*** (0.003)	−0.014*** (0.005)	−0.013*** (0.004)
	Constant	−0.003 (0.009)	−0.001 (0.006)	−0.002 (0.010)	−0.000 (0.010)
Observations		1281	1281	1281	1281

Notes: Outcome is the reported agglomeration elasticity. All models are estimated under PET-PEESE, “precision-effect estimates with standard errors” following Stanley and Doucouliagos (2019), which to correct for publication bias includes a squared standard error term, and its interaction with a developing country dummy, and with a post-1990 dummy, and these terms non-interacted. The standard error terms are not reported for parsimony. PET-PEESE estimates are precision-weighted, using the inverse of the reported estimate’s variance as the weight. Robust standard errors are presented in parentheses, these are clustered at the study level for all model selection methods except Bayesian Model Averaging (BMA). The square brackets reflect the probability of each variable’s model inclusion under BMA. The General model includes all control variables. BIC and AIC are the specifications (from all 17million combinations of controls) that minimize the Bayesian and Akaike Information Criteria respectively. ***, ** and * reflect significance at the 1, 5 and 10 percent levels respectively.

immune from reverse causation and from the effects of more recent shocks affecting both population and urban costs. We include three instruments reflecting the percentage of sand, silt and clay. Unfortunately these instruments are somewhat weaker than historic density, thus these results are presented in the Appendix.

5. Meta-analysis: estimation of agglomeration benefits

This section presents the results of the meta-analysis of agglomeration elasticities. We first examine potential publication bias, present overall meta-estimates and then examine the role of heterogeneity by study characteristics.

a) Publication Bias

The results from Funnel Asymmetry Test (FAT) for publication bias following estimation equation (4), are presented in Table 3. For all estimation methods we find robust evidence of a highly significant positive relationship between the estimated agglomeration elasticity and the corresponding standard error, suggestive of publication bias.

Publication bias can be visualized with a funnel plot of all the reported elasticities against their standard errors (Stanley and Doucouliagos, 2010). The funnel plot in Fig. 5 illustrates that the estimates do not appear to be randomly distributed. More precise estimates (with lower standard errors) are towards the top of the funnel plot and more imprecise are towards the bottom. The presence of selective reporting of positively significant results would mean that the distribution of estimated elasticities should be skewed to the right, i.e. for a given level of estimate precision, more positive elasticities are more likely to be reported. In the absence of selective reporting the estimates should be evenly spread around the true elasticity. In Fig. 5 we include 95 per cent confidence intervals, which form the grey funnel (triangle), around the random effects meta estimate (the red line). In the absence of publication bias, one would expect 95 per cent of elasticity estimates to lie within the grey funnel and to be evenly distributed within it. That is no systematic relationship between reported estimates and their standard errors. Instead reported elasticities are disproportionately skewed to the right, outside the funnel, suggesting positive bias towards reporting statistically significant positive agglomeration elasticities.

We explore a variety of study characteristics to examine whether evidence of publication bias is constrained to a sub-sample of studies, or is more pervasive (following Havranek, 2015). Table 4 presents estimation results of an augmented equation (4), where standard errors are interacted with the study characteristic. Results suggest that publication bias is more severe in developed country than in developing country studies and higher in more recent studies, with a mid-year of data after 1990 (columns 1 and 2). However, publication quality, as measured by the number of citations or publication in a journal, is not significant (columns 3 and 4). At first glance it appears publication bias is more severe for studies using micro-data (column 5), but more recent studies are more likely to use micro-data and also study developing countries. A full specification model with the characteristics together suggest that differences in publication bias are driven by the time period of study and country's income level (column 6).¹³

b) Homogenous meta-estimates

¹³ Interacting the standard error with study characteristics allows testing of significant differences in publication bias. Instead, in Table A3 in the Annex we estimate FAT using different sub-samples of studies, and find evidence of publication bias in both high and non-high-income subsamples, across published and unpublished studies and across studies using both micro and macro-data. Only for the pre-1990 studies do we fail to find evidence of publication bias.

In this section, we present meta-estimates examining agglomeration elasticities across developed and developing economies. To this end, Table 5 presents results from estimating equation (5), with country level income as the only source of heterogeneity in elasticity estimate. Traditional meta-analysis techniques that do not explicitly control for publication bias (columns 1 to 3). Under random effects estimation, we find agglomeration elasticities for developed countries of 5.0 percent, as reflected in the constant term, and for developing countries elasticities are 0.9 percentage points higher (column 1). While not explicitly accounting for publication bias, WLS or fixed effects have been shown to outperform random effects in such cases (Stanley and Doucouliagos, 2014).¹⁴ The WLS or fixed effects find much smaller agglomeration elasticities for developed countries of 1.4 percent and developing countries now 2.2 percentage points higher (columns 2 and 3).

The pervasive evidence of publication bias suggests that the raw meta-estimates can't be taken at their face value. The state-of-the-art methodology, PET-PEESE, that outperforms other estimations under publication bias is presented in columns 4 and 5 of Table 5. We start with PET-PEESE estimation in column 4. Given that the publication bias is stronger for studies post-1990 and differs across studies of developing and developed countries, our preferred specification augments our estimating equation (5) to allow for publication bias that differs across studies based on these variables (column 5).¹⁵ To do so we include interactions of the squared standard error term with these two variables, as well as these terms non-interacted. We find that controlling for publication bias in PET-PEESE does not substantially change the results relative to estimates that don't control for the bias. Agglomeration elasticities appear to be significantly higher than developed countries by around 2.1 percentage points (column 4 and 5).¹⁶ Thus although we found substantial evidence of publication bias in the previous section and that precision weighting (e.g. WLS) matters for estimated elasticities, explicitly accounting for publication bias does not appear to materially affect the estimated agglomeration elasticity premium in developing countries.¹⁷

c) Heterogeneity by study characteristics

Considering the wide range of differences within and across studies on characteristics pertaining to estimation methods, sector of study, skills of workers spatial scale, underlying data and so on, we augment our meta-analysis by controlling for a broad set of study characteristics outlined in Table 1.

Table 6 presents our preferred PET-PEESE specification where we deploy a variety of model selection methods to objectively arrive at the

¹⁴ As shown in Stanley and Doucouliagos (2014) the fixed effects estimate is identical to the WLS estimator, only the estimated variances differ.

¹⁵ Our results are robust, although somewhat less precisely estimated, when not allowing for heterogeneous publication bias.

¹⁶ A robustness check using Andrews and Kasy (2019) estimator finds similarly meta estimates are around 2 percentage points higher in non-high income countries. Depending on the choice of cutoff for probability of publication, the estimator finds a mean high-income true elasticity of 1.6 percent to 2.0 percent, compared to 3.8 percent to 4.8 percent for non-high income. The estimator is available at <https://maxkasy.github.io/home/metastudy/>. We employ PET-PEESE as our preferred estimator it is readily adaptable to heterogeneity by study characteristics.

¹⁷ Recall that PET-PEESE estimation includes study standard error terms to account publication bias. In unreported results we find that the study standard errors are uncorrelated with the low-income country dummy. Thus although the study standard errors are correlated with estimated study elasticity (as shown in Table 3 or 4), PET-PEESE estimation shows a similar estimated agglomeration elasticity premium in developing countries as WLS or fixed effects estimation.

Table 7

Pollution cost elasticities and cost levels - across developed and developing countries.

Outcome:	Log PM2.5 Pollution			Log CO2 Emissions		
	(1)	(2)	(3)	(4)	(5)	(6)
Estimation:	OLS	OLS	IV	OLS	OLS	IV
Developing Country	0.211*** (0.068)	0.270*** (0.097)	0.240* (0.140)	−0.490*** (0.119)	−0.490*** (0.121)	−0.259 (0.165)
Log Population Density	0.394*** (0.037)	0.416*** (0.035)	0.478*** (0.048)	−0.508*** (0.083)	−0.508*** (0.117)	−0.520*** (0.121)
Log Population Density * Developing Country		−0.098 (0.121)	−0.140 (0.171)		−0.001 (0.166)	−0.565* (0.315)
First- Stage F Statistic	—	—	92.0	—	—	19.4
Observations	351	351	351	189	189	189

Note: City-level pollution reflects PM2.5 data for 2013 or 2014 for 351 cities (76 in developing countries). City-level CO2 emissions per capita is a cross-section of data with data available of 2010–2015 for 189 cities (120 in developing countries). The top and bottom 1 percent of population density observations are excluded. Non-HIC reflects a dummy variable for developing countries using World Bank country classifications in 2015. Log population density is measured in 2015 and demeaned such that the constant terms are interpreted at the mean city density. All specifications include a control for city-level night lights. IV estimation is presented in columns 3 and 6, where population density in 2015 is instrumented with the history density in 1975. First-stage results are omitted for parsimony. Broadly similar results using geological instruments are presented in Appendix Table A7. Robust standard errors in parentheses, ***, ** and * reflect significance at the 1, 5 and 10 percent levels respectively.

Table 8

Wellbeing and homicide elasticities and cost levels - across developed and developing countries.

Outcome:	Log Subjective Wellbeing			Log Homicides		
	(1)	(2)	(3)	(4)	(5)	(6)
Estimation:	OLS	OLS	IV	OLS	OLS	IV
Developing Country	−0.160*** (0.024)	−0.153*** (0.024)	−0.189*** (0.039)	1.406*** (0.089)	1.525*** (0.087)	2.114*** (0.246)
Log Population Density	−0.064*** (0.017)	−0.083*** (0.018)	−0.048 (0.054)	−0.007 (0.071)	−0.364*** (0.100)	−1.117** (0.474)
Log Population Density * Developing Country		0.033 (0.032)	0.059 (0.082)		0.506*** (0.144)	0.516 (0.487)
First- Stage F Statistic	—	—	18.2	—	—	47.4
Observations	163	163	163	1103	1103	1103

Note: Subjective wellbeing data is an average value for 2014–2018 available from De Neve and Krekel (2020) for 165 cities (106 in developing countries). Homicide rate data reflects an unbalanced panel of 234 cities (63 in developing countries) over the period 2006 to 2016. Regressions involving homicide data include year fixed effects, such that we present pooled cross-sections. Otherwise see notes to Table 7.

sub-set of controls included as meta-regressors.¹⁸ In addition to the general model including all controls, we employ Bayesian Model Averaging (BMA) to examine robustness of our variable selection (Steel, 2020). We report the posterior inclusion probability (PIP) for each variable under BMA in square brackets (see column 2). Our preferred specification, in column 4, involves estimating all 17 million combinations of covariates to choose the model that minimizes the Bayesian Information Criteria (BIC). Note the variables included in our preferred BIC specification, is similar to that proposed by BMA, excluding all variables with a PIP of less than 69 percent and including all those with a PIP of more than 82 percent. We also examine robustness to using the Akaike Information Criteria (AIC) in column 3.¹⁹

Table 6 suggests that the agglomeration premia in developing countries is less than 1 percentage point higher than developed countries, after controlling for not only publication bias, but also differences in time period, agglomeration or productivity measures, estimation methods and underlying data (columns 1 to 4). Across all specifications, the difference between elasticities in developed and developing countries ranges from 0.4 to 0.5 percentage points across all specifications.

¹⁸ Table 6 builds on our preferred estimator, PET-PEESE given by column 5 in Table 5 and adding additional controls as meta-regressors.

¹⁹ As noted earlier, we use BMA as a robustness for model selection, since BMA does not report clustered standard errors – clustering at the study-level can help control for type I errors.

However, in all specifications, the difference in agglomeration premia is not statistically significant. To examine if the small overall differences between agglomeration premia in developing and developed countries is driven by a few outlier countries, we re-estimate the specification in column 4 (BIC) replacing income level dummy with indicator variables for each country. The estimated country coefficients, with the United States as the reference category, are presented in the Appendix Figure A2. While there are differences across individual countries, as expected, we find that most developed and developing countries are clustered around the reference category. The small differences between agglomeration premia across developed and developing countries broadly holds across the countries in our sample, and is not driven by one or two outlier countries.

Table 6 suggests that while the agglomeration elasticities may be comparable across country income groups, many other differences in data or estimation methods across studies matter much more. More specifically, we make the following observations. *First*, the productivity measure matters when it comes to measuring agglomeration premia. Estimated nominal wages or labor productivity premia are much higher than studies using TFP – by 6.0 and 3.6 percentage points respectively (column 4). *Two*, one possible reason for the difference is urban costs. Urban costs increase local prices and the cost of inputs, which feed through into higher nominal wages or labor productivity, however, TFP

Table 9
Congestion elasticities and cost levels - across developed and developing countries.

Outcome:	Log Congestion		
	(1)	(2)	(3)
Estimation:	OLS	OLS	IV
Developing Country	0.249*** (0.053)	0.279*** (0.065)	0.265 (0.184)
Log Population Density	0.096*** (0.035)	0.103*** (0.040)	0.100* (0.056)
Log Population Density * Developing Country		−0.049 (0.080)	−0.028 (0.242)
First- Stage F Statistic	–	–	18.9
Observations	353	353	353

Note: Congestion reflects the annual additional hours spent driving in rush hours, measured in 2018, and contains data for 353 cities (73 in developing countries). Otherwise see notes to Table 7.

elasticities are theoretically independent of urban costs (Combes and Gobillon, 2015).²⁰ Studies that estimate *real* wages or labor productivity – controlling for urban costs – have 4.2 percentage point lower estimates than those that use nominal values.²¹ Accounting for urban costs and labor share explains the majority of the difference we find between nominal wage and labor productivity estimates and TFP.

Three, agglomeration premia differ substantially across industries. The agglomeration premia in services is on average much higher, around 3.7 percentage points higher than estimates encompassing all industries (column 4). Four, controlling for heterogeneity in firm or worker traits and underlying data also yields interesting results. Studies that control for firm size tends to reduce agglomeration premia, likely because larger firms tend to be more productive and hire more skilled workers). More spatially disaggregated studies tend to find around 1.3 percentage point smaller agglomeration benefits, perhaps because of spillovers on surrounding areas. Unskilled workers are less likely to benefit from density, although the strength of the statistical significance depends on the model selection. Unskilled workers have around 0.3 percentage point lower productivity than workers as a whole (column 4). Whereas skilled workers have around 1.3 percentage point higher productivity (column 3). Five, controlling for sorting of skilled workers in cities either at the local level or using the individual's education reduces agglomeration premia by 1.0 and 1.3 percentage points respectively (column 4). Similarly, more productive firms may choose to locate in cities and agglomeration premia may reflect local characteristics that are hard to measure. Employing panel fixed effects leads to lower estimates of around 2.1 percentage points (column 4). Six, controlling for the so-called endogeneity via instrumental variable estimation typically does not materially impact the results, at least conditional on including the other controls above.

Lastly, there are no robust differences for agglomeration premia estimated in higher and lower quality publications, or across different agglomeration measures (although there is evidence localization controls matter).

The studies we include in our meta-analysis reflect a variety of quality of estimation approaches, in part due to the availability of data. Some studies account for endogeneity by including panel fixed effects or instrumental variables, others account for human capital with worker-level controls and others do not. We follow standard meta-analysis practice and account for heterogeneity in data and method with a rich array of control variables. As a robustness we present our main results

focusing only on studies that include higher quality estimation approach – defined as a historic or geological instrumental variable, panel fixed effects or a worker-level human capital control to address sorting.²² We find very similar results to our baseline, with the agglomeration premia being not significantly different in developing countries (see columns 3 and 4 in Appendix Table A4).²³ As discussed in section 3, the number of higher quality estimates is roughly balanced between developed and developing countries (208 and 279 respectively). We also find very similar results to our baseline when simply restricting the sample to worker data (columns 1 and 2 in Table A4).

In the Appendix Table A5 we examine robustness of our main results to alternative estimation approaches and additional sample restrictions. The main findings shown in Table 6 remain intact, that is, the differences in agglomeration premia across developed and developing countries are small, once publication bias and other study differences are considered. It becomes clearer that underlying data or estimation choices matter far more than country-income. In Appendix Table A5, we decompose the panel fixed effects, industry, firm size and local area human capital controls into those separately reflecting spatial, worker and firm data (column 1). In general, we fail to find that these controls matter differentially across data sources. In column 2, we mitigate any residual risk that our results are driven by a few papers with a large number of estimates (although note we cluster standard errors at the study-level in the baseline estimation). To do so we take a single mean estimate per combination of meta controls in each paper, which reduces the sample substantially to only 318 observations. We do not find agglomeration premia differ significantly between developing and developed countries. This confirms the message of our baseline results that differences across country income groups are small, and other data or estimation choices matter far more. Finally, columns 3 to 6 repeats our preferred specification (from Table 6 column 4) but excludes two developed and developing countries with the largest number of estimates, Brazil, China, UK, USA. The results are unchanged.

6. Estimating urban costs

Our meta-analysis in section 5 suggests that urban costs play a major role in explaining differences in agglomeration benefits across studies. Most studies measure “gross” agglomeration benefits of density rather than advantages of locating in crowded, congested and crime prone places that “net” out these benefits. In sum, the evidence base for urban costs is underdeveloped. Using novel data on crime, congestion,

²⁰ Note also that TFP estimates need to be scaled up by the labor share to be comparable with wages or labor productivity.

²¹ This is broadly in line with differences between real wage and nominal wage elasticities within the same papers, e.g. Chauvin et al. (2017) or Faberman and Freedman (2016).

²² We focus only on worker-level studies, since these account for the majority of studies of developing countries.

²³ We do not present the AIC or BIC model selection in this case, because the developing country dummy is dropped from the optimal model.

wellbeing and pollution from hundreds of cities around the world, including those in developing countries, we present estimates of density elasticity of urban costs in Tables 7–9.

We present OLS estimates in columns 1–2 and 4–5 and IV estimates in columns 3 and 6. Note that we have one first stage for each of the endogenous variables, one for contemporaneous population density and another for contemporaneous population density interacted with the developing country dummy. Accordingly, we include two instruments: historic population density and historic density interacted with the developing country dummy. We find the historic instruments are typically strong predictors as suggested by the first-stage F-statistics. Importantly the OLS and IV estimates are generally similar, suggesting any omitted variable bias, at least controlling for city-level night lights is relatively small. Estimation using geological instruments are somewhat weaker and include in the Appendix in Table A7, but the second stage results are again similar to OLS estimates.

Levels of urban costs are significantly higher in developing countries, and this is true across our measures of congestion and crime, lower levels of subjective wellbeing and depending on the specification, pollution, suggested by the positive developing country dummy (and mirroring earlier Fig. 4). In levels, for the average city density in our data, in developed countries 28 percent fewer hours are spent in traffic congestion, pollution is 23 percent lower, wellbeing is 15 percent higher, and the homicide rate is around 3 times lower. In contrast, carbon dioxide emissions are 39 percent higher in developed country cities.²⁴

Although developing countries have higher urban cost levels, their cost elasticities appear to be generally no different to those observed in developed countries. Relative to developed countries, the elasticity of urban costs in developing countries are similar for PM2.5 pollution, carbon emissions, congestion and subjective wellbeing. Our results point to an elasticity of PM2.5 pollution and congestion with respect to city density to be 39 percent and 10 percent respectively overall and not significantly different between developed and developing countries.²⁵ Unlike PM2.5 pollution, carbon dioxide emissions per capita however decrease substantially with density (as also found by Ahlfeldt and Pietrostefani, 2019 for advanced economies), with an elasticity of –51%. Subjective wellbeing decreases 6% with city density, but again this is similar across income groups. Note that our estimates for subjective wellbeing elasticity are large relative to the estimate of 0.4% for OECD countries from Glaeser et al. (2016). This is likely due to availability of our subjective wellbeing data for relatively few cities and the inherent difficulty of comparing these metrics.

In contrast, the elasticity of crime, measured by the homicide rate, with respect to city density is significantly higher at +14 percent in developing countries compared with –36 percent in developed countries. The latter suggests that if crime is accounted for in measuring the advantages of locating agglomerations, the net benefits in developing countries would be much smaller. Although there is little research in this area, one reason could be that the efficiency of law enforcement does not improve as fast with city density in developing countries - for instance, because of higher incentives for corruption with density, poorer complementary infrastructure, tighter fiscal environment or poor government capabilities.

As a robustness exercise we use city population density from census

data (instead of build-up density) for each country.²⁶ The results are presented in the Appendix Table A6. Although we find somewhat lower urban cost elasticities for PM2.5 pollution (15 percent overall) and CO2 emissions (14 percent) and higher for congestion (21 percent), the key messages do not change. We find that the levels of congestion, crime and pollution are larger in developing economies, but the urban cost elasticities are not discernibly different between developing and developed countries.

7. Monetary equivalents

In this section we bring together the urban cost and agglomeration elasticities and assign approximate monetary equivalents to aide comparability to illustrate the effects from a 1% increase in density. Clearly, this exercise is a back-of-an-envelope calculation, particularly because of limited data on monetary values for developing countries; we draw upon Ahlfeldt and Pietrostefani's (2019) similar exercise for advanced economies. The present value is calculated as the product of 1% (the increase in density), the elasticity with city density, the base value and the unit value to convert to dollars, which is divided by a 5% discount rate.²⁷ We present a range of estimates using both our estimated elasticities, and those elasticities from Ahlfeldt and Pietrostefani's (2019) which uses advanced economies. Our urban cost analysis has shown that aside from homicides, the elasticities in the benefits and returns to agglomeration are not significantly different in developed and developing economies. Thus we typically use the same elasticity for each country group. Furthermore, this means that the difference in monetary values across developed and developing countries is driven predominantly by differences in the base values and unit values.

For gross wages we take the meta-analysis wage elasticity coefficient from Table 6 (the constant term is negligible), which is around 6 percent. We use the same elasticity in both developed and developing countries. This is somewhat higher than the 4 percent figure used in Ahlfeldt and Pietrostefani's (2019) for advanced economies. The base value we take is average income (USD GDP per capita) using the countries within our meta-analysis sample and the unit value is by necessity 1.²⁸ Using the range of 4–6 percent elasticities, we estimate the present value of higher wages due to a 1 percent increase in density to be \$277 to \$416 in developed and \$74 to \$111 in developing countries.

Our pollution elasticity upper bound is 15 percent, which we take from column 1 of Appendix Table A6; this compares to 13 percent from Ahlfeldt and Pietrostefani (2019)²⁹ which we use as a lower bound. We follow Ahlfeldt and Pietrostefani (2019) and use their base value of rent for developed country cities, and for developing countries we adjust this down by the average USD rent index between developed and developing countries from Numbeo.com³⁰. We assume the same rent-pollution elasticity of 0.3 as in Ahlfeldt and Pietrostefani (2019) for in each country group. The present value of higher pollution due a 1 percent increase in density is -\$90 to -\$104 in developed and -\$42 to -\$48 in developing countries.

For CO2 emission reductions we take the elasticity estimate of 0.51

²⁶ Census data is available for only about half the cities in our sample.

²⁷ We assume a common (USD) discount rate across each income group, but in principle one could assume a higher rate in developing countries to reflect additional risk factors.

²⁸ \$34,666 is the average GDP per capita for developed countries for the average developed country and \$9237 for the average developing country.

²⁹ We use this elasticity rather than those in Table 7, because it uses census population density as in Ahlfeldt and Pietrostefani's (2019), so is more comparable with the unit values we also take from upon Ahlfeldt and Pietrostefani's (2019).

³⁰ Thus we use a base value of rent of \$11500 in developed and \$5351 in developing countries.

²⁴ Since the model is log-linear, these percentages are calculated using the exponential of the reported coefficient on the developing country dummy minus one (using the coefficients estimated in columns 1 and 4).

²⁵ Our estimate is somewhat higher than Ahlfeldt and Pietrostefani (2019) who report the pollution elasticity to be +22 percent, using OLS estimation and OECD data. They also find a congestion elasticity of +8 percent, using a different measure of congestion to our paper, average travel speeds using OECD data. Their meta-analysis of 13 high income country studies of crime elasticities (not focusing on homicides as we do) finds a mean elasticity of –24 percent.

from Table 7 for the top of the range, the elasticity of 0.22 from Ahlfeldt and Pietrostefani (2019) for the lower bound, and combine these with the base quantity from our urban cost data. Since carbon emissions are a global public bad, we use the same \$43 cost per ton of carbon in both country groups as Ahlfeldt and Pietrostefani (2019). This gives a present value of lower CO2 emissions of \$10 to \$23 in developed countries and \$4 to \$9 in developing.

Our subjectifnwellbeing elasticities from Table 8 of 0.06 are much higher than the single estimate available in the literature of 0.004 for OECD countries from Glaeser et al. (2016) that is used by Ahlfeldt and Pietrostefani (2019). We prefer to use 0.004 and combine this with the happiness elasticity of 2 following Ahlfeldt and Pietrostefani (2019), and the average income as noted earlier. This suggests the value of the reduction in wellbeing of around -\$55 in developed economies and -\$15 in developing economies.

For homicide reductions we take the elasticities of 0.36 and -0.53 for developed and developing countries from Table 8, and the base quantity from our urban cost data. Ahlfeldt and Pietrostefani (2019) find a crime reduction elasticity for rich countries of 0.085.³¹ The statistical value of life – our unit value – comes from the OECD (2012).³² This gives a value of homicide reductions of \$2 to \$7 in developed countries and of homicide increases of around -\$9 for developing.

The congestion cost elasticity of 0.10 comes from Table 9, compared to 0.12 in Ahlfeldt and Pietrostefani (2019). We combine these elasticities with the base level of congestion from our urban cost data. We use the hourly wage as the unit value, calculated from the income per capita (see footnote 34) and assuming 260 7-h working days per year. The present value the increase in congestion is -\$41 to -\$49 in developed countries and -\$15 to -\$18 in developing countries.

8. Conclusions

We examine nearly 1300 estimates of agglomeration elasticities from 76 studies covering 34 countries over the period 1973 to 2020. While there are well-known examples in the literature of high agglomeration returns in China, India and countries in Africa, these often mask others such as Chile that have negative elasticity estimates. Using a novel meta-analysis, we find that the elasticity estimates in developing countries are less than 1 percentage point higher than in advanced economies, but not significantly so.

Instead our meta-analysis suggests the wide differences in raw estimates in the literature can be attributed to estimation set-up (outcome and agglomeration measure), consideration of urban costs in estimating net benefits, and the variation in pay-off across sectors and skills. Studies using wages as an outcome variable have higher estimated elasticities compared to those using TFP measures, while studies using population size or density have larger estimates relative to market access measures. Service sector estimates are on average higher than manufacturing, while skilled workers disproportionately benefit from density relative to others.

Appendix

Our most important finding is that nominal wages, the canonical measure of agglomeration economies, seldom capture the costs of working and living in cities. These costs include time lost in commuting as well as dis-amenities such as pollution or crime, and lower subjective wellbeing. Our meta-analysis shows that studies report real returns (net of urban costs) find elasticities to be 4.2 percentage points lower than studies that do not, implying a net agglomeration elasticity of essentially zero (-0.6 percent) for developed countries when using labor productivity as an outcome measure. Although our analysis confirms the relatively higher “gross” wage elasticities with respect to density; we find no evidence for “net” gains in light of high and increasing costs of working in developing country cities.

Our novel estimates of urban dis-amenities suggest that only the elasticity of crime is higher in developing country cities. Although the elasticity of pollution and congestion in developing countries is comparable with developed countries, their levels are much higher. Higher costs in developing country cities are partly due to bad design and lack of capital investment, but also by the fact that their growth is not driven by the process of structural transformation, which would create a critical mass of more productive firms that benefit from sharing, matching, and learning. Many developing country cities are not dense and productive—they are just crowded.³³

Author statement

Jonathan Timmis: Conceptualization, Data Curation, Methodology, Formal analysis, Software, Writing - Original Draft, Writing - Review & Editing, Visualization. Arti Grover: Conceptualization, Methodology, Writing - Original Draft, Writing - Review & Editing. Somik Lall: Conceptualization, Writing - Review & Editing.

Declaration of competing interest

The authors (Arti Grover, Somik Lall and Jonathan Timmis) have no competing interests to declare.

Data availability

Data will be made available on request.

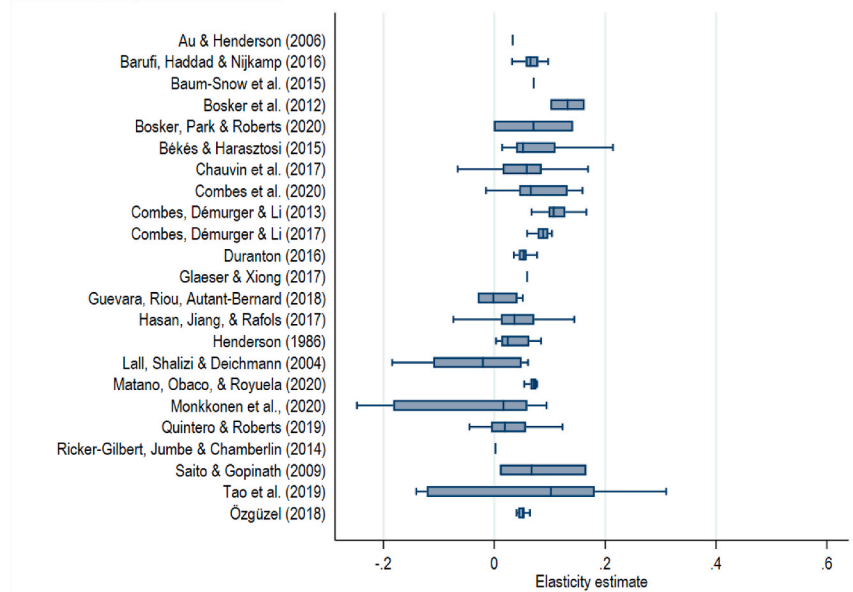
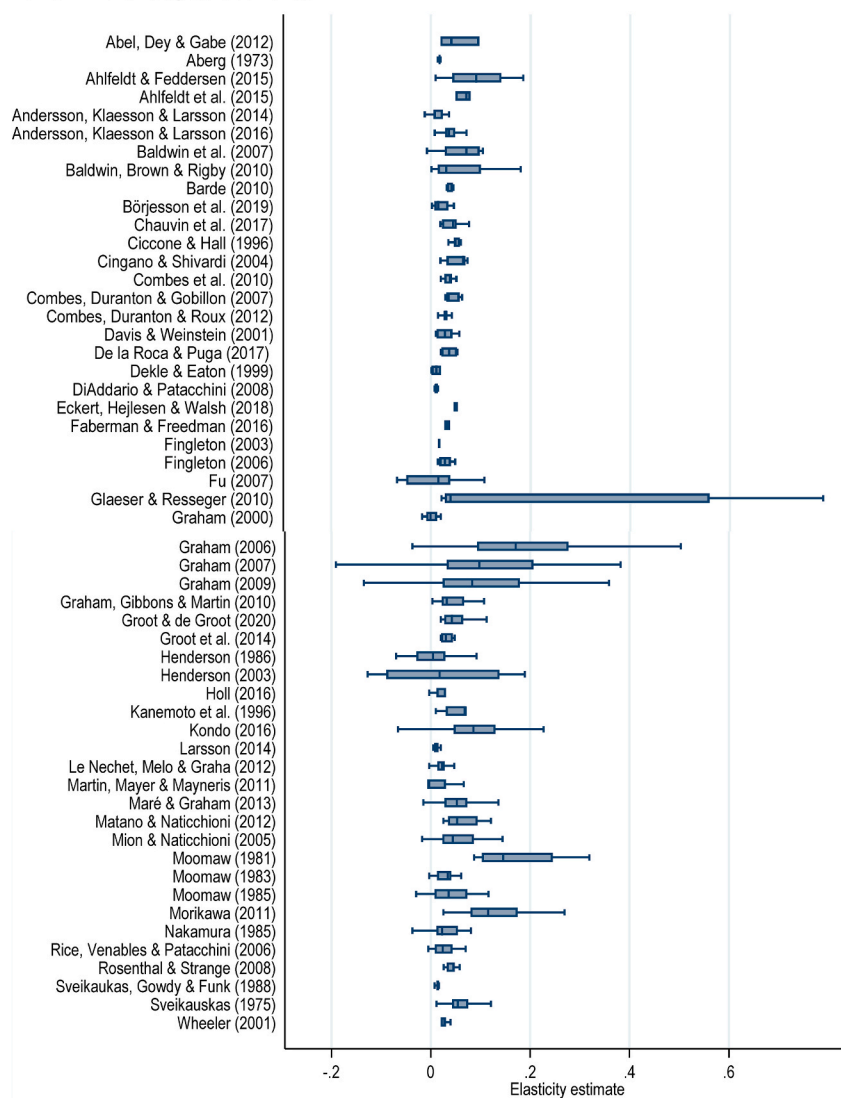
Acknowledgements

We thank Gilles Duranton and Bill Maloney for insightful discussions; Somya Bajaj, Andrii Berdnyk and Yewon Choi for excellent research assistance; Mark Roberts, Luis Quintero, Patricia C. Melo, Daniel J. Graham, Robert B. Noland, Gabriel M. Ahlfeldt, and Elisabetta Pietrostefani for generously sharing the underlying data for their meta-analysis studies, Nancy Lozano Gracia, Olivia D’aoust, Maria Edisa Soppelsa, Hogeun Park, Katie McWilliams, for pointing us to complementary data sources on urban costs.

³¹ We do not apply this to developing countries because our urban cost analysis shows homicide elasticities differ across these income groups.

³² OECD (2012) proposes a value of \$4million per life in OECD economies. They find an income elasticity of 0.8, which suggests a value of \$1.4million using the average GDP for our developing country sample (see footnote 34).

³³ Note see the full list of studies included in the meta-analysis in Table A8.

Panel A – Developing Countries**Panel B – Developed Countries****Fig. A1.** Elasticity estimates by Study

Notes: The box shows the 25th, median and 75th percentile elasticity for each paper, with the whiskers showing the upper and lower adjacent values. Some papers

contain a single estimate, hence only the median is shown.

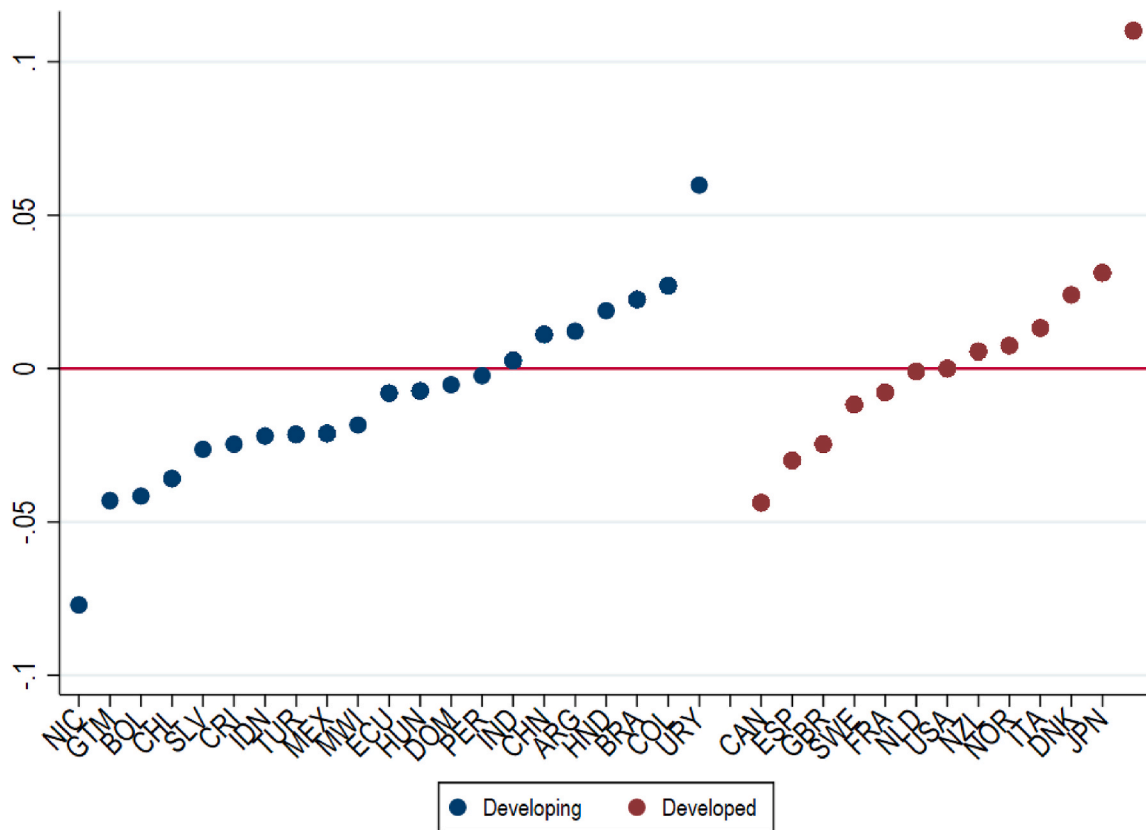


Fig. A2. Meta-Analysis Estimated Country Agglomeration Elasticity (vs USA)

Notes: Estimated under PET-PEESE, “precision-effect estimates with standard errors” following [Stanley and Doucouliagos \(2019\)](#), which to correct for publication bias includes a squared standard error term, and its interaction with a developing country dummy, and with a post-1990 dummy. Model selection under Bayesian Information Criterion, equivalent of column 4 in [Table 6](#), but allowing for separate country dummies in place of a developing country dummy. USA is the omitted reference category, represented by the horizontal line at zero. ***, ** and * reflect significance at the 1, 5 and 10 percent levels respectively.

Table A1
Mean agglomeration elasticities by meta variable category

Category	Variable	Estimated Elasticity	
		Mean of reference category (dummy = 0)	Mean if dummy = 1
Country Income	Developing Country	0.057	0.060
Productivity Measure	Wages	0.063	0.051
	Labor Productivity	0.060	0.044
Urban Cost	Accounts for Urban Costs	0.058	0.034
Industry	Manufacturing Sector	0.061	0.052
	Services Sector	0.049	0.109
Skill	Skilled workers	0.058	0.059
	Unskilled workers	0.059	0.027
Time Period	Post-1990s	0.037	0.063
Study Quality	Published	0.074	0.052
Spatial Measure	City - Level	0.069	0.048
	Sub City - Level	0.052	0.071
Agglomeration Measure	Density Measure	0.056	0.059
	Market Potential Measure	0.053	0.073
Localization	Localization Control	0.062	0.047
Data	Panel Data	0.053	0.061
	Firm Data	0.048	0.077
	Worker Data	0.059	0.056
Endogeneity	Panel Fixed Effects	0.061	0.046
	IV estimation - contemporaneous	0.061	0.012
	IV estimation - historic	0.058	0.057
Firm Heterogeneity	Industry control	0.059	0.054
	Firm size control	0.054	0.064
Worker Sorting	Local area human capital	0.063	0.033
	Individual-level human capital	0.056	0.062

Note: Number of observations for each variable is equal to 1281. Table shows the mean agglomeration elasticity for each of the dummy variable categories listed in Table 1.

Table A2

Summary statistics of controls included in meta-analysis – by Developed and Developing Country Samples. Notes: See notes to Table 1 for further details on the variable construction. 376 Developing country estimates and 905 Developed country estimates.

Category	Variable	Developing		Developed	
		mean	std dev	mean	std dev
Country Income	Non-High Income Country	1.000	0.00	0.00	0.00
Productivity Measure	Wages	0.787	0.410	0.335	0.472
	Labor Productivity	0.043	0.202	0.217	0.412
Urban Cost	Accounts for Urban Costs	0.051	0.219	0.023	0.151
Industry	Manufacturing Sector	0.261	0.440	0.444	0.497
	Services Sector	0.154	0.362	0.149	0.356
Skill	Skilled workers	0.032	0.176	0.015	0.123
	Unskilled workers	0.013	0.115	0.031	0.173
Time Period	Post-1990	0.947	0.225	0.723	0.448
Study Quality	Published	0.785	0.412	0.716	0.451
	Number of Citations	−0.452	1.325	0.188	1.326
Spatial Measure	City - Level	0.277	0.448	0.645	0.479
	Sub City - Level	0.285	0.452	0.312	0.463
Agglomeration Measure	Density Measure	0.707	0.456	0.343	0.475
	Market Potential Measure	0.040	0.196	0.336	0.473
Localization	Localization Control	0.207	0.406	0.343	0.475
Data	Panel Data	0.495	0.501	0.636	0.481
	Firm Data	0.120	0.325	0.433	0.496
	Worker Data	0.763	0.426	0.259	0.438
Endogeneity	Panel Fixed Effects	0.120	0.325	0.248	0.432
	IV estimation - contemporaneous	0.035	0.183	0.081	0.272
	IV estimation - historic	0.378	0.485	0.083	0.276
Firm Heterogeneity	Industry control	0.362	0.481	0.213	0.410
	Firm size control	0.311	0.464	0.397	0.489
Worker Sorting	Local area human capital	0.114	0.319	0.212	0.409
	Individual-level human capital	0.630	0.483	0.128	0.334

Table A3

Funnel Asymmetry Test for Publication Bias Across Data Subsamples

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Estimation Method:	WLS							
Sub Sample:	Developing Countries	Developed Countries	Post-1990	Pre-1990	Published	Unpublished	Micro Data	Macro Data
Standard error	1.857*** (0.533)	4.430*** (0.714)	4.678*** (0.581)	−0.349 (1.090)	3.096*** (0.415)	3.538*** (1.077)	4.589*** (0.664)	1.968*** (0.527)
Constant	0.026** (0.011)	0.011*** (0.004)	0.009*** (0.002)	0.038** (0.014)	0.011*** (0.004)	0.029*** (0.008)	0.012*** (0.004)	0.010 (0.006)
Observations	376	905	1010	271	943	338	958	323

Notes: Estimation of equation (4). WLS are precision-weighted least squares estimates, using the inverse of the reported estimate's variances as weights. Robust standard errors are presented in parentheses. Reported agglomeration elasticity expressed as absolute values. Standard errors are clustered at the study level. The estimation is conducted over different subsamples of the data. ***, ** and * reflect significance at the 1, 5 and 10 percent levels respectively.

Table A4

Meta results robustness – restricting to Worker Data, and Worker Data With Higher Quality Estimation Techniques

		(1)	(2)	(3)	(4)
		PET-PEESE			
		Worker-Data		Worker-Data Higher Quality Studies	
		General	BMA	General	BMA
Country Income	Developing Country	0.005 (0.006)	0.000 (0.002)	0.003 (0.006)	0.000 (0.001)
Urban Cost	Accounts for Urban Costs	−0.011* (0.005)	−0.000 (0.004)	−0.010* (0.005)	−0.001 (0.004)
Industry	Manufacturing Sector	0.024*** (0.005)	0.024*** (0.005)	0.023*** (0.004)	0.021*** (0.005)
	Services Sector	0.042*** (0.006)	0.036 (0.022)	0.041*** (0.006)	0.037** (0.018)
Skill	Skilled workers	0.012*** (0.004)	0.003 (0.006)	0.011*** (0.004)	0.005 (0.007)
	Unskilled workers	−0.002*** (0.000)	−0.001 (0.001)	−0.002*** (0.000)	−0.000 (0.001)

(continued on next page)

Table A4 (continued)

		(1)	(2)	(3)	(4)
Estimation Method:		PET-PEESE			
Model Selection:		Worker-Data		Worker-Data Higher Quality Studies	
		General	BMA	General	BMA
Time Period	Post-1990	0.000 (0.006)	0.000 (0.001)	0.001 (0.007)	−0.000 (0.001)
Study Quality	Published	−0.004 (0.005)	−0.008* (0.004)	−0.004 (0.005)	−0.002 (0.004)
	Number of Citations	−0.006** (0.003)	−0.004 (0.003)	−0.003* (0.002)	−0.002 (0.002)
Spatial Measure	City - Level	−0.000 (0.001)	−0.000 (0.000)	−0.000 (0.000)	0.000 (0.000)
	Sub City - Level	−0.017** (0.006)	−0.014** (0.006)	−0.015*** (0.005)	−0.014*** (0.005)
Agglomeration Measure	Density Measure	−0.001 (0.006)	0.000 (0.002)	0.001 (0.007)	−0.000 (0.001)
	Market Potential Measure	−0.004 (0.007)	−0.003 (0.005)	−0.012* (0.007)	−0.016*** (0.004)
Localization	Localization Control	0.006 (0.003)	0.001 (0.003)	0.002 (0.003)	0.000 (0.001)
Data	Panel Data	−0.006 (0.005)	−0.007 (0.006)	−0.007 (0.005)	−0.003 (0.005)
Endogeneity	Panel Fixed Effects	−0.036*** (0.006)	−0.038*** (0.003)	N/a	N/a
	IV estimation - contemporaneous	−0.007*** (0.003)	−0.002 (0.004)	N/a	N/a
	IV estimation - historic	0.005 (0.005)	0.003 (0.003)	N/a	N/a
Firm Heterogeneity	Industry control	0.013*** (0.003)	0.012*** (0.003)	0.006* (0.004)	0.009*** (0.003)
	Firm size control	−0.002*** (0.000)	−0.000 (0.001)	−0.002*** (0.000)	−0.001 (0.001)
Worker Sorting	Local area human capital	−0.021*** (0.007)	−0.022*** (0.006)	N/a	N/a
	Individual-level human capital	−0.028*** (0.005)	−0.029*** (0.003)	N/a	N/a
	Constant	0.064*** (0.008)	0.067*** (0.006)	0.033*** (0.005)	0.029*** (0.005)
	Observations	521	521	487	487

Notes: Outcome is the reported agglomeration elasticity. Columns 1 to 6 are estimated under PET-PEESE, “precision-effect estimates with standard errors” following Stanley and Doucouliagos (2019), which to correct for publication bias includes a squared standard error term, and its interaction with a developing country dummy, and with a post-1990 dummy. The standard error terms are not reported for parsimony. All estimates are precision-weighted, using the inverse of the reported estimate’s variance as the weight. All models are variations of the baseline BIC estimation of column 4 of Table 6 (model selection by Bayesian Information Criterion). In columns 1 and 2 we restrict the sample to estimates that only use worker data and exclude meta-controls that are no longer relevant (e.g. firm data, worker data). In columns 3 and 4 we further restrict the data to studies employing higher quality estimation methods (historic or geological instrumental variable, panel fixed effects or a worker-level human capital control). Standard errors are clustered at the study-level. ***, ** and * reflect significance at the 1, 5 and 10 percent levels respectively.

Table A5

Meta results robustness to alternative estimation approaches and sample restrictions

Sample Restrictions		(1)	(2)	(3)	(4)	(5)	(6)
		Full Sample – Additional FEs	Single Estimate	Exclude GBR	Exclude USA	Exclude BRA	Exclude CHN
Country Income	Developing	0.006 (0.007)	−0.007 (0.006)	0.002 (0.008)	0.005 (0.008)	−0.005 (0.009)	0.005 (0.008)
Productivity Measure	Wages	0.033*** (0.008)	0.044*** (0.007)	0.065*** (0.013)	0.066*** (0.017)	0.067*** (0.014)	0.062*** (0.013)
	Labor Productivity	0.022** (0.009)	0.027*** (0.009)	0.034** (0.013)	0.037** (0.014)	0.039*** (0.012)	0.039*** (0.013)
Urban Cost	Accounts for Urban Costs	−0.035*** (0.009)	−0.018** (0.009)	−0.050*** (0.014)	−0.046*** (0.015)	−0.042*** (0.014)	−0.043*** (0.013)
Industry	Manufacturing Sector	0.011** (0.005)	−0.005 (0.007)	0.020*** (0.007)	0.018** (0.008)	0.014** (0.007)	0.017** (0.007)
	Services Sector	0.037*** (0.011)	0.041*** (0.005)	0.061*** (0.012)	0.037*** (0.011)	0.036*** (0.011)	0.035*** (0.011)
Skill	Unskilled workers	−0.003*** (0.001)	−0.002*** (0.001)	−0.003*** (0.001)	−0.003*** (0.001)	−0.003** (0.001)	−0.003** (0.001)
Time Period	Post-1990s	0.008 (0.007)	0.000 (0.004)	−0.010 (0.006)	−0.004 (0.010)	−0.000 (0.006)	−0.003 (0.006)
Study Quality	Published	−0.007 (0.006)	−0.013*** (0.004)	−0.015** (0.007)	−0.006 (0.006)	−0.005 (0.005)	−0.005 (0.005)
Spatial Measure	Sub City - Level	−0.013***	−0.007**	−0.007**	−0.013***	−0.014***	−0.013***

(continued on next page)

Table A5 (continued)

Sample Restrictions		(1)	(2)	(3)	(4)	(5)	(6)
		Full Sample – Additional FEs	Single Estimate	Exclude GBR	Exclude USA	Exclude BRA	Exclude CHN
Agglomeration Measure	Market Potential Measure	(0.003) –0.005 (0.005)	(0.003) –0.003 (0.004)	(0.003) –0.001 (0.007)	(0.003) –0.005 (0.008)	(0.003) –0.006 (0.007)	(0.003) –0.006 (0.007)
	Localization Control	–0.002 (0.004)	0.000 (0.004)	0.001 (0.004)	0.004 (0.007)	0.002 (0.005)	0.003 (0.005)
Data	Firm Data	0.038*** (0.009)	0.033*** (0.005)	0.049*** (0.013)	0.048*** (0.017)	0.047*** (0.012)	0.046*** (0.011)
	Worker Data	0.011 (0.007)	–0.016** (0.006)	–0.013 (0.008)	–0.017* (0.009)	–0.020** (0.008)	–0.016* (0.009)
Endogeneity	Panel Fixed Effects	–0.001 (0.004)	–0.019*** (0.006)	–0.024*** (0.006)	–0.020*** (0.007)	–0.022*** (0.006)	–0.021*** (0.006)
	Panel Fixed Effects * Firm Data	0.004 (0.015)					
	Panel Fixed Effects * Worker Data	–0.036*** (0.007)					
Firm Heterogeneity	Industry control	0.005 (0.009)	0.008*** (0.003)	0.006** (0.003)	0.010** (0.004)	0.012*** (0.003)	0.011*** (0.003)
	Industry control * Firm Data	–0.009 (0.010)					
	Industry control * Worker Data	0.008 (0.009)					
	Firm size control	0.010 (0.011)	–0.003*** (0.000)	–0.003** (0.002)	–0.004** (0.002)	–0.004* (0.002)	–0.004* (0.002)
	Firm size control * Firm Data	–0.025** (0.012)					
	Firm size control * Worker Data	–0.012 (0.011)					
	Constant	0.006 (0.009)	0.015** (0.006)	0.008 (0.009)	–0.004 (0.015)	–0.003 (0.010)	–0.003 (0.010)
Observations		1281	318	1106	1066	1168	1196

Notes: Outcome is the reported agglomeration elasticity. Columns 1 to 6 are estimated under PET-PEESE, “precision-effect estimates with standard errors” following Stanley and Doucouliagos (2019), which to correct for publication bias includes a squared standard error term, and its interaction with a developing country dummy, and with a post-1990 dummy. The standard error terms are not reported for parsimony. All estimates are precision-weighted, using the inverse of the reported estimate’s variance as the weight. All models are variations of the baseline BIC estimation of column 4 of Table 6 (model selection by Bayesian Information Criterion). Full Sample – Additional FEs decomposes the panel fixed effects, industry, firm size and local area human capital controls into those separately reflecting spatial, worker and firm data. Single Estimate uses a single mean estimate per combination of meta controls in each paper. Columns 3 to 6 repeat the baseline estimation but exclude the two developed and developing countries with the largest number of estimates – UK, US, Brazil and China respectively. Standard errors are clustered at the study-level. ***, ** and * reflect significance at the 1, 5 and 10 percent levels respectively.

Table A6

Using census population data, there urban cost elasticities are not statistically different between developed and developing countries.

Panel A: Pollution and Wellbeing Elasticities						
	Log PM2.5 Pollution		Log CO2 Emissions		Subjective Wellbeing	
	(1)	(2)	(3)	(4)	(5)	(6)
Developing Country	0.613*** (0.112)	0.608*** (0.112)	–0.935*** (0.133)	–0.932*** (0.133)	–0.187*** (0.034)	–0.187*** (0.034)
Log Population Density	0.153*** (0.033)	0.208*** (0.037)	–0.142* (0.083)	–0.188 (0.132)	–0.057** (0.022)	–0.058** (0.022)
Log Population Density * Developing Country		–0.107* (0.057)		0.081 (0.161)		0.003 (0.048)
Observations	161	161	69	69	55	55
Panel B: Crime and Congestion Elasticities.						
	Log Homicides		Log Congestion			
	(1)	(2)	(3)	(4)		
Developing Country	1.174*** (0.096)	1.174*** (0.096)	0.316*** (0.040)	0.318*** (0.040)		
Log Population Density	0.064 (0.063)	0.054 (0.064)	0.209*** (0.024)	0.236 (0.032)		

(continued on next page)

Table A6 (continued)

Panel B: Crime and Congestion Elasticities.			
	Log Homicides		Log Congestion
	(1)	(2)	(3) (4)
Log Population Density * Developing Country		0.020 (0.125)	–0.060 (0.042)
Observations	584	584	153 153

Notes: Population density reflects population per square km from census data. The top and bottom 1percent of population density observations are excluded. Otherwise see notes to [Tables 7–9](#)

Table A7

Congestion elasticities and cost level – Geological Instruments

Outcome:	Log PM2.5 Pollution	Log Homicides	Log Congestion
	(1)	(2)	(3)
IV - Geological			
Developing Country	0.116 (0.189)	2.545*** (0.492)	0.253 (0.262)
Log Population Density	0.751*** (0.122)	–0.964 (0.596)	0.251** (0.098)
Log Population Density * Developing Country	–0.327 (0.263)	–1.197 (1.343)	–0.202 (0.345)
First- Stage F Statistic	5.0	3.4	3.4
Observations	351	1103	353

Note: Presents IV estimation using three geological instruments of sub-soil percentage of silt, sand and clay. Other outcomes are not presented due to very weak instruments. Otherwise see notes for [Tables 7–9](#)

Meta-Analysis Methodology

Meta-Analysis Paper Selection

The selection of papers is driven by different search approaches. Several combinations of keywords were used to find papers in EconLit, ScienceDirect and Google Scholar. The search was completed on 13th April 2020.

The keywords were used to reflect agglomeration measures, productivity measures and country income. All of the following keywords were used individually: agglomeration, elasticity, density, urban, cost, developing countries, developed countries, productivity, urban, wages, output. In addition, the following keyword combinations were employed:

- Agglomeration, elasticity, density
- Agglomeration, elasticity, urban
- Agglomeration, density, cost
- Agglomeration, density, urban
- Agglomeration, elasticity, productivity
- Agglomeration, elasticity, output
- Urban, cost, density
- Urban, wages, density
- Agglomeration, elasticity, developing country
- Agglomeration, elasticity, developed country
- Density, elasticity, developing country
- Density, elasticity, developed country

Given the vast results on EconLit, ScienceDirect and Google Scholar, we limited our search to the first 15 pages, where the pages are sorted based on the relevance of the studies for the keyword combination. Since the focus of our work is on developing countries where estimations of agglomeration elasticity are quite recent, we only focused on studies that were published after 2002. For prior works, we rely on the underlying data for meta-analysis provided in [Melo et al. \(2009\)](#).

We complemented this approach with analysis of citation trees for the papers that inspired the study ([Melo et al. \(2009\)](#), [Chauvin et al. \(2017\)](#) and [Ahlfeldt and Pietrostefani \(2019\)](#)), as well as publications and working papers noted on the websites of the most cited authors (Pierre-Philippe Combes, Gilles Duranton, Edward Glaeser, Laurent Gobillon, Stephen Redding, Esteban-Rossi Hansberg, Daniel J. Graham). Lastly, we consulted with colleagues at the World Bank and academic institutions to identify any ongoing work in the field.

The initial sample included peer-reviewed papers, working papers of universities or specialist research institutes (World Bank, NBER, CEPR, CESifo, and IZA), chapters in books and conference proceedings.

Estimates were excluded that did not report standard errors (due to our correction for publication bias), those with economic scale outcomes rather than productivity (note we include production function approaches), or those not reporting estimates by individual countries (e.g. by region, such as [Henderson et al., 2019](#)).

Some papers report semi-elasticities (from a log-linear model) that we convert to an elasticity. Two papers ([Combes et al., 2012](#); [Kondo, 2016](#))

report estimated agglomeration premia at above and below average densities. These have been converted to an elasticity by taking the difference in agglomeration premia divided by the difference in densities at these two points, similar to the meta-analysis of [Ahlfeldt and Pietrostefani \(2019\)](#). Studies are excluded that cannot be represented as an elasticity.

The list of variables collected for each study is given by [Table 1](#). The literature review was conducted by Somya Bajaj, and reviewed by Jonathan Timmis. The resulting sample is 1281 estimates from 76 papers from 1973 to 2020.

Table A8

Full List of Studies

#	Authors	Journal	Volume, Page No.	Paper Name	Countries
1	Abel et al. (2012)	Journal of Regional Science	52 (4), pp. 562-586	Productivity and the density of human capital	USA
2	Aberg (1973)	Regional and Urban Economics	3 (2), pp. 131-155	Regional productivity differences in Swedish manufacturing	SWE
3	Ahlfeldt & Feddersen (2015)	Journal of Economic Geography	111, pp. 93-107	From periphery to core - Measuring agglomeration effects using high-speed rail	DEU
4	Ahlfeldt et al. (2015)	Econometrica	83 (6), pp. 2127-2189	The economics of density - evidence from the Berlin wall	DEU
5	Andersson, Klaesson & Larsson (2014)	Regional Science	93 (4), pp. 727-747	The sources of the urban wage premium by worker skills - spatial sorting or agglomeration economies?	SWE
6	Andersson, Klaesson & Larsson (2016)	Regional Studies	50 (6), pp. 1082-1095	How local are spatial density externalities?	SWE
7	Au and Henderson (2006)	The Review of Economic Studies	73 (3), pp. 549-576	Neighborhood effects in agglomeration economies	CHN
8	Baldwin et al. (2007)	Economic Analysis (EA) Research Paper Series	No. 2007045	Urban economies and productivity	CAN
9	Baldwin, Brown & Rigby (2010)	Journal of Regional Science	50 (5), pp. 915-934	Agglomeration economies - Microdata panel estimates from Canadian manufacturing	CAN
10	Barde (2010)	Spatial Economic Analysis	5 (1), pp. 73-91	Increasing returns and the spatial structure of French wages	FRA
11	Barufi, Haddad & Nijkamp (2016)	Annals of Regional Science	56, pp. 707-755	Industrial scope of agglomeration economies in Brazil	BRA
12	Baum-Snow et al. (2015)	European Regional Science Association Conference Papers	ersa15p1177	Transport infrastructure, urban growth and market access in China	CHN
13	Békés and Harasztosi (2013)	Regional Science and Urban Economics	43 (1), pp. 51-64	Agglomeration premium and trading activity of firms	HUN
14	Börjesson et al. (2019)	Economics of Transportation	18, pp. 27-39	Agglomeration, productivity and the role of transport system improvements	SWE
15	Bosker et al. (2012)	Journal of Urban Economics	72 (2-3), pp. 252-266	Relaxing Hukou - Increased labor mobility and China's economic geography	CHN
16	Bosker, Park & Roberts (2020)	Journal of Urban Economics	In press	Definition matters - metropolitan areas and agglomeration economies in a large developing country	IDN
17	Carlsen, Ratts and Stokke (2016)	Regional Science and Urban Economics	60, pp. 39-49	Education, experience, and urban wage premium	NOR
18	Chauvin et al. (2017)	Journal of Urban Economics	98, pp. 17-49	What is different about urbanization in rich and poor countries? Cities in Brazil, China, India and US	BRA, CHN, IND, USA
19	Ciccone and Hall (1996)	The American Economic Review	86 (1), pp. 141-154.	Productivity and the density of economic activity	USA
20	Cingano and Shivardi (2004)	Journal of the European Economic Association	2(4), pp. 720-744.	Identifying the sources of local productivity growth	ITA
21	Combes et al. (2019)	Journal of Development Economics	142	Unequal migration and urbanisation gains in China	CHN
22	Combes et al. (2010)	National Bureau of Economic Research Chapter	In Agglomeration Economics, pp. 15-66	Estimating agglomeration economies with history, geology, and worker effects	FRA
23	Combes et al. (2013)	CEPR Discussion Paper	No. 9352	Urbanisation and Migration Externalities in China	CHN
24	Combes, Démurger & Li (2017)	Universite de Lyon Working Paper	Working Papers 1709	Productivity gains from agglomeration and migration in Chinese cities over 2002-2013	CHN
25	Combes et al. (2008)	Journal of Urban Economics	63 (2), pp. 723-742	Spatial wage disparities - sorting matters	FRA
26	Combes et al. (2012)	Econometrica	80, pp. 2543-2594	The productivity advantages of large city: distinguishing agglomeration from firm selection	FRA
27	Davis and Weinstein (2001)	NBER Working paper	No. 8518	Market size, linkages, and productivity - a study of Japanese regions.	JPN
28	De la Roca and Puga (2017)	Review of Economic Studies	84 (1), pp. 106-142	Learning by Working in Big Cities	ESP
29	Dekle and Eaton (1999)	Journal of Urban Economics	46 (2), pp. 200-214	Agglomeration and land rents - evidence from the prefectures	JPN
30	DiAddario and Patacchini (2008)	Labour Economics	15 (5), pp. 1040-1061	Wages and the City: Evidence from Italy	ITA
31	Duranton (2016)	Journal of Regional Science	56 (2), pp. 210-38	Agglomeration effects in Colombia	COL
32	Eckert, Hejlesen & Walsh (2018)	Opportunity and Inclusive Growth Institute Working Paper	No. 24	The return to big city experience - evidence from danish refugees	DNK
33	Faberman and Freedman (2016)	Journal of Urban Economics	93, pp. 71-84	The urban density premium across establishments	USA
34	Fingleton (2003)	Oxford Economic Papers	55 (4), pp. 716-739	Increasing returns - evidence from local wage rates in Great Britain	GBR

(continued on next page)

Table A8 (continued)

#	Authors	Journal	Volume, Page No.	Paper Name	Countries
35	Fingleton (2006)	Oxford Economic Papers	58 (3), pp. 501-530	The new economic geography versus urban economics - an evaluation using local wage rates in Great Britain	GBR
36	Fu (2007)	Journal of Urban Economics	61 (1), pp. 86-111	Smart café city- testing human capital externalities in the Boston metropolitan area	USA
37	Glaeser & Resseger (2010)	Journal of Regional Science	50 (1), pp. 221-244	The complementarity between cities and skills	USA
38	Glaeser & Xiong (2017)	Oxford Review of Economic Policy	33 (3), pp. 373-404	Urban productivity in the developing world	CHN
39	Graham (2000)	International Review of Applied Economics	14 (3), pp. 323-341	Spatial variation in labor productivity in British manufacturing	GBR
40	Graham (2006)	Transportation Research Board 85th Annual Meeting	06-0531	Transport investment, agglomeration and urban productivity	GBR
41	Graham (2007)	Journal of Transport Economics And Policy	41 (3), pp. 317-343	Agglomeration, productivity and transport investment	GBR
42	Graham (2009)	Papers in Regional Science	88 (1), pp. 63-84	Identifying urbanization and localization externalities in manufacturing and service industries	GBR
43	Graham et al. (2010)	LSE Working Paper	October 2010	The spatial decay of agglomeration economies - estimates for use in transport appraisal	GBR
44	Groot and de Groot (2020)	De Economist	168 (1), pp. 53-78	Estimating the skill bias in agglomeration externalities and social returns to education - evidence from Dutch matched worker-firm micro-data	NLD
45	Groot et al. (2014)	Journal of regional science	54 (3), pp. 503-523	Regional wage differences in the Netherlands - micro evidence on agglomeration externalities	NLD
46	Guevara, Riou, Autant-Bernard (2018)	University of Lyon Working Paper	No. 1818	Agglomeration externalities in Ecuador - do urbanisation and tertiarisation matter?	ECU
47	Hasan and Rafols (2017)	Asian Development Review	34 (2), pp. 201-228	Urban agglomeration effects in India: evidence from town-level data	IND
48	Henderson (1986)	Journal of Urban Economics	19 (1), pp. 47-70	Efficiency of resource usage and city size	BRA, USA
49	Henderson (2003)	Journal of Urban Economics	53 (1), pp. 1-28	Marshall's scale economies	USA
50	Holl (2016)	Journal of Urban Economics	93, pp. 131-151	Highways and productivity in manufacturing firms	ESP
51	Kanemoto et al. (1996)	Journal of the Japanese and International Economies	10 (4), pp. 379-398	Agglomeration economies and a test for optimal city sizes in Japan	JPN
52	Kondo (2016)	REIT Discussion Paper	No. 16098	Testing for agglomeration economies and firm selection in spatial productivity differences: the case of Japan	JPN
53	Lall, Shalizi & Deichmann (2004)	Journal of Development Economics	73 (2), pp. 643-673	Agglomeration economies and productivity in Indian industry	IND
54	Larsson (2014)	The Annals of Regional Science	52, pp. 367-384	The neighborhood or the region? reassessing the density-wage relationship using geocoded data	SWE
55	Le Nechet, Melo & Graha (2012)	Transportation Research Record	2307 (1), pp. 21-30	Transportation-induced agglomeration effects and productivity of firms in megacity region of Paris basin	FRA
56	Maré and Graham (2013)	Journal of Urban Economics	75, pp. 44-56	Agglomeration elasticity and firm heterogeneity	FRA
57	Martin, Mayer & Mayneris (2011)	Journal of Urban Economics	69 (2), pp. 182-195	Spatial concentration and plant-level productivity in France	NZL
58	Matano and Naticchioni (2012)	Journal of Regional Science	56 (2), pp. 191-209	What drives the urban wage premium? Evidence along the wage distribution	ITA
59	Matano, Obaco & Royuela (2020)	Journal of Regional Science	60(4), pp. 823-847	What drives the spatial wage premium in formal and informal labor markets? The case of Ecuador	ECU
60	Mion and Naticchioni (2009)	CEPR Discussion Papers	No. 5172	Urbanization externalities, market potential and spatial sorting of skills and firms	ITA
61	Monkkonen et al. (2020)	Urban studies	57 (10), pp. 2080-2097	Compact city and economic productivity in Mexico	MEX
62	Moomaw (1981)	The Quarterly Journal of Economics	96 (4), pp. 675-688	Productivity and city size - a critique of the evidence	USA
63	Moomaw (1983)	Regional Science and Urban Economics	13, pp. 525-545	Is population scale a worthless surrogate for business agglomeration economies?	USA
64	Moomaw (1985)	Journal of Urban Economics	17 (1), pp. 73-89	Firm location and city size - reduced productivity advantages as a factor in the decline of manufacturing in urban areas	USA
65	Morikawa (2011)	The Review of Economics and Statistics	93 (1), pp. 179-192	Economies of density and productivity in service industries - an analysis of personal service industries based on establishment-level data	JPN
66	Nakamura (1985)	Journal of Urban Economics	17 (1), pp. 108-124	Agglomeration economies in urban manufacturing industries - a case of Japanese cities	JPN
67	Özgüzel (2018)	Economic Research Forum Working Papers	No. 1341	Agglomeration effects in a developing economy - evidence from Turkey	TUR
68	Quintero and Roberts (2018)	World Bank Policy Research Paper	No. 8560	Evidence from 16 Latin American and Caribbean countries	ARG, BOL, BRA, CHL, COL, CRI, DOM, ECU, GTM, HND, MEX, NIC, PER, SLV, URY
69	Rice et al. (2006)	Regional Science and Urban Economics	36 (6), pp. 727-752	Spatial determinants of productivity - analysis for the regions of Great Britain	GBR
70	Ricker-Gilbert, Jumbe & Chamberlin (2014)	Food Policy	48, pp. 114-128	How does population density influence agricultural intensification and productivity? evidence from Malawi	MWI

(continued on next page)

Table A8 (continued)

#	Authors	Journal	Volume, Page No.	Paper Name	Countries
71	Rosenthal et al. (2008)	Journal of Urban Economics	64 (2), pp. 373-389	The attenuation of human capital spillovers	USA
72	Saito and Gopinath (2009)	Journal of Economic Geography	9 (4), pp. 539-558	Plants self-selection, agglomeration economies and regional productivity in Chile	CHL
73	Sveikauskas, Gowdy & Funk (1988)	Economics of Transportation,	28 (2), pp. 185-202	Urban productivity- city size or industry size	USA
74	Sveikauskas (1975)	The Quarterly Journal of Economics	89 (3), pp. 393-413	The productivity of cities	USA
75	Tao et al. (2019)	Regional Science and Urban Economics	77, pp. 141-154	Agglomeration economies in creative industries	CHN
76	Wheeler (2001)	Journal of Labor Economics	19 (4), pp. 879 - 899	Search, sorting, and urban agglomeration	USA

References

- Abel, Dey & Gabe, 2012. Productivity and the density of human capital. *J. Reg. Sci.* 52 (4), 562–586.
- Ades, A.F., Glaeser, E., 1995. Trade and circuses: explaining urban giants. *Q. J. Econ.* 110 (1), 195–227.
- Ahlfeldt, G.M., Pietrostefani, E., 2019. The economic effects of density: a synthesis. *J. Urban Econ.* 111 (C), 93–107. Elsevier.
- Ahlfeldt, et al., 2015. The economics of density - evidence from the Berlin wall. *Econometrica* 83 (6), 2127–2189.
- Akbar P., V. Couture, G. Duranton and A. Storeygard (forthcoming). Mobility and Congestion in Urban India. *American Econ. Rev.*
- Andrews, I., Kasy, M., 2019. Identification of and correction for publication bias. *Am. Econ. Rev.* 109 (8), 2766–2794.
- Arzaghi, M., Henderson, J.V., 2008. Networking off madison avenue. *Rev. Econ. Stud.* 75, 1011–1038.
- Avner, P., Lall, S.V., 2016. Matchmaking in Nairobi: the Role of Land Use" World Bank Policy Research Working Paper No. 7904. World Bank.
- Bacolod, M., Blum, B.S., Strange, W.C., 2009. Urban interactions: soft skills vs. specialization. *J. Econ. Geogr.* 9, 227–262.
- Baum-Snow, N., Pavan, R., 2011. Understanding the city size wage gap. *Rev. Econ. Stud.* 79 (1), 88–127.
- Baum-Snow, N., Brandt, L., Henderson, J.V., Turner, M.A., Zhang, Q., 2017. Roads, railroads, and decentralization of Chinese cities. *Rev. Econ. Stat.* 99 (3), 435–448.
- Bernard, Louise, 2016. Job Access in Kampala. University of Oxford.
- Card, D., Kreuger, A., 1995. Time-series minimum-wage studies: a meta-analysis. *Am. Econ. Rev.* 85 (2), 238–243.
- Carozzi, F., Roth, S., 2023. Dirty density: air quality and the density of American cities. *J. Environ. Econ. Manag.* 118, 102767.
- Castells-Quintana, D., Dienesch, E., Krausne, M., 2021. Air pollution in an urban world: a global view on density, cities and emissions. *Ecol. Econ.* 189, 107153.
- Chauvin, J.P., Glaeser, E., Ma, Y., Tobio, K., 2017. What is different about urbanization in rich and poor countries? Cities in Brazil, China, India and the United States. *J. Urban Econ.* 98, 17–49.
- Ciccone, A., Hall, R.E., 1996. Productivity and the density of economic activity. *Am. Econ. Rev.* 86 (1), 54–70.
- Combes, P.P., Gobillon, L., 2015. The empirics of agglomeration economies. In: Duranton, G., Henderson, J.V., Strange, W.C. (Eds.), *Handbook of Regional and Urban Economics*, vol. 5. Elsevier, North-Holland, pp. 247–348.
- Combes, P.P., Duranton, G., Gobillon, L., 2008. Spatial wage disparities: sorting matters. *J. Urban Econ.* 63 (2), 723–742.
- Combes, P.P., Duranton, G., Gobillon, L., Puga, D., Roux, S., 2010. Estimating agglomeration effects with history, geology, and worker fixed-effects. In: Glaeser, E.L. (Ed.), *Agglomeration Economics*, vol. 1. Chicago University Press, Chicago, IL, pp. 15–65.
- Combes, P.P., Duranton, G., Gobillon, L., Roux, S., 2012. The productivity advantages of large cities: distinguishing agglomeration from firm selection. *Econometrica* 80, 2543–2594.
- Combes, P.P., Duranton, G., Gobillon, L., 2013. The costs of agglomeration: land prices in French cities. In: PSE Working Papers Halshs-00849078. HAL.
- Combes, P.P., Duranton, G., Gobillon, L., 2019. The costs of agglomeration: house and land prices in French cities. *Rev. Econ. Stud.* 86 (4), 1556–1589.
- D'Costa, S., Overman, H., 2014. The urban wage growth premium: sorting or learning? *Reg. Sci. Urban Econ.* 48 (C), 168–179.
- De la Roca, J., Puga, D., 2017. Learning by working in big cities. *Rev. Econ. Stud.* 84 (1), 106–142.
- De Luca, G., Magnus, J., 2011. Bayesian model averaging and weighted-average least squares: equivariance, stability, and numerical issues. *STATA J.* 11 (4), 518–544.
- De Neve, Krekel, 2020. Cities and Happiness: A Global Ranking and Analysis. World Happiness Report.
- De Soto, H., 2000. *The Mystery of Capital: Why Capitalism Triumphs in the West and Fails Everywhere Else*. Basic Books, New York.
- Desmet, K., Fafchamps, M., 2006. Employment concentration across U.S. counties. In: *Regional Science and Urban Economics*, vol. 36. Elsevier, pp. 482–509, 4.
- DiAddario, S., Patacchini, E., 2008. Wages and the city: evidence from Italy. *Lab. Econ.* 15 (5), 1040–1061.
- Duranton, G., 2008. Viewpoint: from cities to productivity and growth in developing countries. *Can. J. Econ.* 41 (3), 689–736.
- Duranton, G., 2015. Growing through cities in developing countries. *World Bank Res. Obs.* 30 (1), 39–73.
- Duranton, G., 2016. Agglomeration effects in Colombia. *J. Reg. Sci.* 56 (2), 210–238.
- Duranton, G., Puga, D., 2004. Chapter 48 - micro-foundations of urban agglomeration economies. In: Henderson, J.V., Thisse, J.-F. (Eds.), *Handbook of Regional and Urban Economics*, vol. 4, pp. 2063–2117.
- Duranton, G., Puga, D., 2020. The economics of urban density. *J. Econ. Perspect.* 34 (3), 3–26.
- Duranton, G., Ghani, E., Grover, A., Kerr, W., 2015. The misallocation of land and other factors of production in India. In: Policy Research Working Paper No. 7221. World Bank Group, Washington, DC.
- Egger, M., Smith, G.D., Schneider, M., Minder, C., 1997. Bias in meta-analysis detected by a simple, graphical test. *BMJ* 315, 629.
- Ellis, P., Roberts, M., 2016. Leveraging Urbanization in South Asia: Managing Spatial Transformation for Prosperity and Livability. World Bank, Washington D.C.
- Faberman, R.J., Freedman, M., 2016. The urban density premium across establishments. *J. Urban Econ.* 93 (C), 71–84. Elsevier.
- Florax, R., 2001. Methodological Pitfalls in Meta-Analysis: Publication Bias, Serie Research Memoranda 0028. VU University Amsterdam, Faculty of Economics, Business Administration and Econometrics.
- Fragoso, T., Bertoli, W., Louzada, F., 2018. Bayesian model averaging: a systematic review and conceptual classification. *Int. Stat. Rev.* 86 (1), 1–28.
- Fujita, Masahisa, Ogawa, H., 1982. Multiple equilibria and structural transition of non-monocentric urban configurations. *Reg. Sci. Urban Econ.* 12 (2), 161–196.
- Fujita, M., Krugman, P., Mori, T., 1999. On the evolution of hierarchical urban systems. *Eur. Econ. Rev.* 43 (2), 209–251.
- Gaubert, C., 2018. Firm sorting and agglomeration. *Am. Econ. Rev.* 108 (11), 3117–3153.
- Ghani, E., Grover, A., Kerr, W.R., 2016. Highway to success: the impact of the golden quadrilateral project for the location and performance of Indian manufacturing. *Econ. J.* 126 (591), 317–357.
- Glaeser, E.L., Maré, D.C., 2001. Cities and skills. *J. Labor Econ.* 19 (2), 316–342.
- Glaeser, E.L., Poterba, J.M., 2020. Economic analysis and infrastructure investment. In: NBER Working Paper 28215. National Bureau of Economic Research, Cambridge, MA.
- Glaeser, E.L., Gottlieb, J.D., Ziv, O., 2016. Unhappy cities. *J. Labor Econ.* 34 (2), S129–S182.
- Gollin, D., Jedwab, R., Vollrath, D., 2016. Urbanization with and without industrialization. *J. Econ. Growth* 21 (1), 35–70.
- Graham, D., Gibbons, S., Martin, R., 2010. The spatial decay of agglomeration economies - estimates for use in transport appraisal. In: LSE Working Paper.
- Greenstone, M., He, G., Li, S., Zou, E., 2021. China's war on pollution: evidence from the first five years. In: NBER Working Paper W28467.
- Groot, S., de Groot, H., 2020. Estimating the skill bias in agglomeration externalities and social returns to education - evidence from Dutch matched worker-firm micro-data. *Economist* 168 (1), 53–78.
- Grover, A., Lall, S., Maloney, W., 2022. Place, Productivity, and Prosperity : Revisiting Spatially Targeted Policies for Regional Development. World Bank.
- Harris, C., 1954. The market as a factor in the localization of industry in the United States. *Annals of Association of American Geography* 44, 315–348.
- Hasan, R., Jiang, Y., Rafols, R.M., 2017. Urban agglomeration effects in India: evidence from town-level data. *Asian Dev. Rev. Stud. Asian Pac. Econ. Issues* 34 (2), 201–228.
- Havranek, T., 2015. Measuring intertemporal substitution – the importance of method choices and selective reporting. *J. Eur. Econ. Assoc.* 13 (6), 1180–1204.
- Havranek, T., Rusnak, M., Sokolova, A., 2017. Habit formation in consumption: a meta-analysis. *Eur. Econ. Rev.* 95, 142–167.
- Henderson, J.V., 1974. The sizes and types of cities. *Am. Econ. Rev.* 64, 640–656.
- Henderson, J.V., 2002. Urbanization in developing countries. *World Bank Res. Obs.* 17 (1), 89–112.
- Henderson, J.V., 2003. Marshall's scale economies. *J. Urban Econ.* 53, 1–28.
- Henderson, J. Vernon, Tim Squires, Adam, Storeygard, Weil, David, 2018. The global distribution of economic activity: nature, history, and the role of trade. *Q. J. Econ.* 133 (1), 357–406.

- Henderson, J.V., Nigmatulina, D., Kriticos, S., 2019. Measuring urban economic density. *J. Urban Econ.* (in press).
- Jedwab, R., Moradi, A., 2012. Revolutionizing transport: modern infrastructure, agriculture and development in Ghana. In: *Modern and Comparative Seminar*, 8th March 2012, London School of Economics and Political Science.
- Jedwab, R., Ianchovichina, E., Haslop, F., 2022. Consumption cities versus production cities : new considerations and evidence (English). In: *Policy Research Working Paper ; No. WPS 10105*. World Bank Group, Washington, D.C.
- Kondo, K., 2016. Testing for Agglomeration Economies and Firm Selection in Spatial Productivity Differences: the Case of Japan. Discussion papers 16098. Research Institute of Economy, Trade and Industry (RIETI).
- Lall, S., Henderson, J.V., Venables, A.J., 2017. Africa's Cities: Opening Doors to the World. World Bank, Washington, DC.
- Lindley, J., Machin, S., 2014. Spatial changes in labour market inequality. *J. Urban Econ.* 79 (C), 121–138. Elsevier.
- Melo, P.C., Graham, D.J., Noland, R.B., 2009. A meta-analysis of estimates of urban agglomeration economies. *Reg. Sci. Urban Econ.* 39, 332–342.
- Mion and Naticchioni, 2009. The spatial sorting and matching of skills and firms. *Can. J. Econ.* 42 (1), 28–55.
- Nangini, C., et al., 2019. A global dataset of CO2 emissions and ancillary data related to emissions for 343 cities. *Sci. Data* 6, 180280.
- OECD, 2012. *Mortality Risk Valuation in Environment, Health and Transport Policies*. <https://doi.org/10.1787/9789264130807-en>.
- Peralta-Quirós, T., 2015. *Mobility for All: Getting the Right Urban Indicator*. Connections; No. 25. World Bank, Washington, DC.
- Quintero, L.E., Roberts, M., 2018. Explaining spatial variations in productivity: evidence from Latin America and the caribbean. In: *Policy Research Working Paper*. World Bank. No. 8560.
- Rice, P., Venables, A.J., Patacchini, E., 2006. Spatial determinants of productivity: analysis for the regions of great britain. *Reg. Sci. Urban Econ.* 36 (6), 727–752.
- Rosenthal, S., Strange, W., 2003. Geography, industrial organization, and agglomeration. *Rev. Econ. Stat.* 85, 377–393.
- Rosenthal, S.S., Strange, W.C., 2004. Evidence on the nature and sources of agglomeration economies. In: Henderson, J.V., cois Thisse, J.F. (Eds.), *Handbook of Regional and Urban Economics*, vol. 1. Elsevier, North-Holland, pp. 2119–2171.
- Rosenthal, S.S., Strange, W.C., 2008. The attenuation of human capital spillovers. *J. Urban Econ.* 64, 373–389.
- Stanley, T.D., 2008. Meta-regression Methods for Detecting and Estimating Empirical Effect in the Presence of Publication Bias, vol. 70. *Oxford Bulletin of Economics and Statistics*, pp. 103–127.
- Stanley, T.D., et al., 2013. Meta-analysis of Economics Research Reporting Guidelines. *Journal of Economic Surveys* 27 (2), 390–394.
- Stanley, T.D., Doucouliagos, H., 2010. Picture this: a simple graph that reveals much ado about research. *J. Econ. Surv.* 24 (1), 170–191.
- Stanley, T.D., Doucouliagos, H., 2014. Meta-regression approximations to reduce publication selection bias. *Res. Synth. Methods* 5, 60–67.
- Stanley, T.D., Doucouliagos, H., 2015. Neither fixed nor random: weighted least squares meta-analysis. *Stat. Med.* 34, 2116–2127.
- Stanley, T.D., Doucouliagos, H., 2019. Practical significance, meta-analysis and the credibility of economics. In: *IZA Discussion Paper*. No. 12458.
- Steel, M., 2020. Model averaging and its use in economics. *J. Econ. Lit.* 58 (3), 644–719.
- Storeygard, A., 2016. Farther on down the road: transport costs, trade and urban growth in sub-saharan Africa. *Rev. Econ. Stud.* 83 (3), 1263–1295.
- Tao, J., Ho, C.-Y., Luo, S., Sheng, Y., 2019. Agglomeration economies in creative Industries. *Reg. Sci. Urban Econ.* 77, 141–154.
- United Nations, 2018. *World Urbanization Prospects*. Department of Economic and Social Affairs Population Division, New York.
- van Erven, T., Grunwald, P., de Rooij, S., 2012. Catching up faster by switching sooner: a predictive approach to adaptive estimation with an application to the AIC–BIC dilemma (with discussion). *J. Roy. Stat. Soc.* 74, 361–417.
- Venables, A.J., 2017. Breaking into tradables: urban form and urban function in a developing city. *J. Urban Econ.* 98 (C), 88–97. Elsevier.
- World Bank, 2020. *Convergence: Five Critical Steps toward Integrating Lagging and Leading Areas in the Middle East and North Africa*. World Bank, Washington DC.